Carbon Emissions are Underpriced Due to the Snow Albedo Feedback Effect*

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Abstract

To balance the tradeoff between economic activity and climate goals, a social cost of carbon is estimated to approximate the actual cost of emissions. In this study, we examine changes in surface albedo over time which remain unmodeled in much of the carbon pricing literature. To this end, we combine a state-space model with recent satellite data to estimate incoming energy less albedo from 1850-2023. The results show a decreasing albedo trend from 1940-2023. Moreover, our estimates are consistent with the snow albedo feedback (SAF) where melting ice leads to declines in albedo, thereby causing increased heat absorption at the surface and further melting. Together, these findings indicate that the heating effects of carbon emissions are amplified by the SAF, the omission of which in the carbon pricing decision leads to underpricing.

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*The authors are grateful... The usual caveat applies.

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1 Introduction

The primary sources of energy emit carbon (e.g., crude oil and natural gas) thereby causing most production processes to emit some amount of carbon (Alajingi and Marimuthu, 2023). A common response to this issue among developed economies comes through carbon taxes or tradable pollution permits which require a price of carbon emissions (Jenkins, 2014), and therefore the question of how to price carbon emissions is at the center of contemporary climate and economic policy discussion (Baranzini et al., 2017). The goal of this paper is to consider to what extent the snow albedo feedback (SAF) amplifies temperature effects of carbon emissions which would change their price.

Integrated assessment models (IAMs) are used to project future carbon emissions in response to climate policies and incorporate interactions between climate, social and economic factors to evaluate social costs of emissions (Weyant, 2017). These calculations are used to price carbon emissions in order to internalize the pollution costs. IAMs use a constant climate sensitivity parameter to show how much the global average temperature will rise when the quantity of CO2 in the atmosphere doubles. This climate sensitivity parameter does not capture the SAF because it is forced to be a constant linear function of temperature (Calel and Stainforth, 2017).

Surface albedo refers to the fraction of incoming energy from the sun that reaches the surface of the Earth and is reflected. In this study, albedo refers to surface albedo unless top-of-atmosphere (TOA) albedo (i.e., the fraction of energy reflected by both the atmosphere and surface) is explicitly mentioned. Albedo is important in the climate literature because change in albedo has been theorized to amplify climate change through a positive feedback loop between temperature and frozen surfaces, referred to as the SAF. Specifically, the SAF mechanism indicates that for every degree the planet heats up, some ice melts, which exposes less reflective surfaces and leads to increased heat absorption at the surface. This leads to subsequent melting and heating. Because carbon emissions contribute to global temperature increases, the SAF could amplify heating effects of carbon emissions. Recent estimates from Kaufmann and Pretis (2023) suggest that the direct effect of carbon emissions on temperature is amplified by 3.4% due to SAF, which motivates further research on the topic.

To this end, our study provides evidence that surface albedo has declined from 1940-2023 in a manner consistent with SAF theory. Our model improves on past studies of surface albedo because we leverage time series methods for estimating unobserved variables combined with commonly used satellite data to consider a longer pre-satellite era timespan. Specifically, we embed a benchmark energy balance model (EBM) from Brock and Miller (2024) in a state space model (SSM) and estimate incoming radiation less albedo as an unobserved state variable using the Kalman filter and MLE. Additionally, we propose a modification to the Kalman filter/MLE procedure which treats the state variable as partially observed and allows use of modern satellite data from 2001-

2023.

Our primary specification shows that albedo changes from 1940-2023 account for a 0.9 W m^{-2} increase in the Earth surface energy budget, approximately 50% of the direct effect of carbon emissions over that time. This finding suggests that the SAF amplification of carbon emissions is large enough to cause underpricing in IAMs.

2 Literature Review

The term albedo was first coined by Swiss mathematician Johann Heinrich Lambert referring to the fraction of light radiated back from a surface in optics (Mach, 2013; Perkins, 2020), but it was not until the American astronomer George P. Bond that it was used to study the reflectiveness of planets (Bond, 1861). In this seminal work, Bond postulated that surface and atmospheric factors shape the reflection of astronomical objects, both of which form the basis for studying planetary albedo to this day. Therefore, planetary albedo is the fraction of incoming solar energy reflected by the Earth back to space. Planetary albedo represents the direct relationship between the outgoing flux at the top of the atmosphere and the incoming solar flux at the top of the atmosphere (Stephens et al., 2015). The terms TOA albedo and planetary albedo are used interchangeably in the literature (Stephens et al., 2015; Liang et al., 2019; Zhan and Liang, 2022; Nikolov and Zeller, 2024). In contrast, surface albedo is the percentage of incoming solar radiation that reaches the surface of the Earth and is reflected (Riihelä et al., 2024) thereby serving as a contributor to planetary albedo together with atmospheric scattering (Stephens et al., 2015) as previously noted by Bond (1861).

Current albedo estimates in climate science typically rely on satellite systems dedicated to the analysis of the Earth radiation budget such as the Earth Radiation Budget Experiment (ERBE) and the Clouds and the Earth's Radiant Energy System (CERES) and therefore are restricted to the post-satellite era. Regarding TOA albedo estimates, one of the most widely used sources is CERES[1] which provides incoming flux and outgoing flux measurements dating back to 2000. CERES TOA fluxes are used by Stephens et al. (2015) and Hansen et al. (2025) to estimate TOA albedo. Given TOA observations from satellites, several algorithms have been used to calculate surface albedo. For instance, Li and Garand (1994) estimate surface albedo from ERBE TOA albedo using a formula depending on solar zenith angle and total precipitable water which they show to be robust across several surface types. Similarly, He et al. (2014) compares observations from nine satellite-based surface albedo data sets which convert satellite observations to surface albedo estimates using multiple algorithms for removing the effects of atmosphere and cloud cover. CERES produces the most accurate TOA fluxes (Zhan et al., 2019), and researchers generate alternative TOA products to improve the spatio-temporal resolution rather than accuracy of the 1-

[1] https://ceres.larc.nasa.gov/data/

degree spatial, monthly aggregated values provided by CERES (Song et al., 2018; Zhan et al., 2019; Zhan and Liang, 2022). Beyond this data collection, the CERES team combines TOA observations with separate data sets (e.g., regarding cloud properties, temperature, humidity) and conducts a calibration whereby surface, cloud and atmospheric properties are adjusted until computed TOA fluxes match observed TOA fluxes.

Although satellites have enabled accurate measurement of albedo, these methods are limited to times when satellites have been in the sky. Even the longest satellite albedo series only extends back to 1979[2]. Before the use of satellites, one of the most reliable methods to measure TOA albedo was earthshine methods which infer the reflectiveness of the Earth based on observed brightness of the moon. According to the historical review by Stephens et al. (2015), early work with this method published in 1913, 1928 and 1933 generated TOA albedo estimates of 89%, 29% and 39%, respectively, with such large variations in estimates over a short period of time indicating much lower precision compared to later estimates. The earliest earthshine observations by Very (1913) and Danjon (1928, 1933) focused on obtaining TOA albedo measurements (Hunt et al., 1986) and at best would only give information on surface albedo based on assumptions of atmospheric and cloud conditions at the time the estimates were made.

While earthshine methods remain in use during the satellite era (Goode et al., 2001; Pallé et al., 2004; Palle et al., 2016; Goode et al., 2021), several issues have been noted with this method including Loeb et al. (2007) who disputes large variations in TOA albedo found by Pallé et al. (2004) which contrasted with TOA albedo estimates from CERES satellite data. Regarding the sampling of Pallé et al. (2004), Loeb notes that the earthshine observations only observe 1/3 of the Earth and 1/3 of the lunar cycle at any given time. This contrasts with TOA CERES measurements which sample a broader and more consistent view of the reflectiveness of the planet. Another issue occurred with problematic conversions of the earthshine measurements obtained from a surface observatory to TOA fluxes (Stephens et al., 2015). Furthermore, Wu et al. (2023) has noted that even among earthshine measurement methods, the technology to collect this data has improved over time bringing into question the earliest earthshine estimates conducted by Very (1913) and Danjon (1928, 1933). Finally, even among the studies producing earthshine time series, the data series do not extend for very long and the ones that do come from the past were collected very sporadically (Goode et al., 2001; Stephens et al., 2015; Goode et al., 2021) hindering our ability to generate long time series from these data to use in long-term climate and economics studies.

In addition to earthshine methods, Stephens et al. (2015) mention methods of calculating TOA albedo relying on radiation budget calculations with assumptions of cloud cover and surface char-

[2]CM SAF Cloud, Albedo and surface Radiation (CLARA) climate data record which provides albedo estimates from 1979-2020 (Riihelä et al., 2024) primarily based on narrowband AVHRR (Advanced Very High Resolution Radiometer) measurements.

acteristics, again with estimates varying extensively. According to their review, the earliest of these estimates comes from Abbott and Fowle (1908) who estimated planetary albedo by separating the planet into subsections by type of cloud cover and computing radiation movements. One hurdle Abbott and Fowle (1908) faced was lacking precise estimates of various parameters such as surface and cloud albedo. Therefore, while this study was an important step forward, modern instruments like satellites were necessitated to get information about climate characteristics. Building on this, later estimates from the 1940's-1960's (Fritz, 1949; Houghton, 1954) used similar climate modeling to estimate planetary albedo and were able to leverage improved climate parameter estimates due to advances in data collection technology and sampling frequency. However, even these relatively more modern studies of the pre-satellite age had difficulty in estimating TOA albedo and had an even less clear picture regarding surface albedo. Given the difficulty of estimating albedo without satellite observations, to the best of our knowledge no data series exists for albedo estimates extending from the pre-satellite era continuously to the near-present.

The lack of reliable pre-satellite era albedo estimates presents issues related to the identification of long-term cycles in albedo and could subsequently bias causal estimates of forcing on albedo and temperature. For instance, Kosaka and Xie (2013) suggest that a long-term cycle explains the 1998-2013 hiatus in global warming which some have argued was proof that the link between anthropogenic forcing and temperature was insignificant. Similarly, as-yet-unidentified cycles present in global mean albedo could mean current understandings of the forcing-albedo and forcing-temperature relationships may be faulty. Moreover, the satellite-era albedo estimates are infeasible to incorporate in long-term studies of the forcing-temperature relationship such as that done by Brock and Miller (2024) due to the restricted time frame of available data. For this reason, our study aims to extend the time frame under consideration by combining satellite albedo estimates with time series methods to back out the properties of pre-satellite era albedo which to the best of our knowledge has not been done before on a global scale[3].

Although limited to recent time periods, satellite technology has enabled researchers to study changes in albedo across time with improved precision. For instance, Kato (2009) uses satellite data from 2000-2004 to show global mean TOA albedo exhibits only small changes between years. Likewise, Stephens et al. (2015) surveys the literature on global TOA albedo and finds that estimates of TOA albedo have not changed much since the invention of satellites in 1959. Moreover, Stephens et al. (2015) find that the small observed variations in TOA flux can be explained by cloud cover changes as opposed to differences in surface reflection. Particularly, Stephens et al. (2015) uses CERES TOA data from 2000-2013 and finds no evidence of changes in TOA albedo. However,

[3]Previous research from Song et al. (2021) combined current satellite observations with surface albedo look-up maps and reanalysis data to extract surface albedo dating back to 850 AD in China, but the current study expands on this by providing global average surface albedo which is useful for broader climate applications.

Hansen et al. (2025) analyses the same data product from 2000-2024 and finds accelerated declines in TOA albedo starting around 2015. Goode et al. (2021) supports the Hansen et al. (2025) estimates with both earthshine measurements from the Big Bear Solar Observatory and CERES satellite data. Specifically, the earthshine estimates of TOA albedo decline from 1998-2017 and correspond to a forcing of approximately 0.5 watts per meter squared (W m^{-2}), while the CERES data from 2001-2017 show a TOA albedo decrease corresponding to a forcing of 1.5 W m^{-2} with the authors specifically noting a sharp decline in the CERES data starting in 2015 which is beyond the last year that Stephens et al. (2015) covers. Moreover, Nikolov and Zeller (2024) substantiate this trend further using CERES data and claim that global warming can largely be explained by large decreases in TOA albedo documented over this time frame.

Regarding surface albedo variation, Mengyao et al. (2023) examined nine different land surface albedo data sets over the available time spans (some starting in ∼2000 and others ∼1980) and found no long-term trend in land surface albedo. However, Mengyao et al. (2023), Riihelä et al. (2024) and Qu et al. (2015) have noted that commonly available surface albedo datasets have difficulty estimating the albedo of ocean surfaces, sea ice and ice over polar regions, all of which are of primary concern in modern theory on albedo changes over time given the SAF (Previdi et al., 2021; Kaufmann and Pretis, 2023). For this reason, Mengyao et al. (2023) restrict their trend analysis to land surface albedo and therefore it is possible the study does not capture variation in albedo due to receding ice lines. Along these lines, Diebold and Rudebusch (2022) use time series methods to project how fast Arctic sea ice will melt in the future and find that in recent years, Arctic sea ice extent has been decreasing at an increasing rate. Specifically, Diebold and Rudebusch (2022) fit a quadratic model to Arctic sea ice data in order to account for climate feedbacks. Using this statistical model, Diebold and Rudebusch (2022) project a 60% probability of a September ice-free Arctic appearing sometime in the 2030's, which is concerning given that this projection is much earlier than CMIP5 models estimate. Similarly, Diebold and Rudebusch (2023) emphasize the importance of accurately modeling receding ice lines by showing commonly used climate models underestimate the effect of carbon emissions on Arctic sea ice extent. Specifically, Diebold and Rudebusch (2023) fit ice coverage as a linear function of cumulative carbon emissions and find a much higher sensitivity of ice to carbon emissions than models from both CMIP5 and CMIP6. Given that ice is among the most reflective surfaces on Earth, both studies provide evidence that albedo is decreasing over time. Together, the above studies show disagreement in the literature regarding if albedo is truly stable across time.

Given the lack of consensus regarding how much albedo has changed in recent years, we next examine possible sources of albedo shifts across time, beginning with dark materials covering snow and ice such as black carbon (Forster et al., 2024) or algae (Williamson et al., 2020). Regarding black carbon on snow and ice, Kang et al. (2020) find that black carbon accumulation is responsible for about 20% of the drop in albedo in the Tibetan plateau region and point to the necessity for further work in examining the impact of such darkening of existing snow and ice on global albedo trends. Surface albedo declines from black carbon on snow are also supported by a laboratory experiment conducted by Hadley and Kirchstetter (2012) comparing the albedo of snow with and without black carbon accumulation. The albedo declines due to black carbon from this experiment are consistent with both model and field-based estimates. However, black carbon from anthropogenic activity is not the only source of snow and ice darkening as biological organisms can also cause similar albedo declines. Supporting this, field sampling conducted by Williamson et al. (2020) provides evidence that algae blooms increase heat absorption of the Greenland ice sheet and account for a large fraction of albedo variability in the region according to Moderate Resolution Imaging Spectroradiometer (MODIS) data.

Next, land use forms a significant anthropogenic contributor to surface albedo changes in recent history through actions such as deforestation and construction of human habitats. Specifically, urbanization has been shown to both decrease (Tang et al., 2018) and increase (Hu et al., 2016) albedo depending on type of surface change, and deforestation has also been shown to increase albedo given that forests absorb much light (Mylne and Rowntree, 1992; Santos Orozco et al., 2023). Along these lines, Ouyang et al. (2022) analyze the global warming effect of urbanization from 2001-2018 on albedo and find a planetary heating effect, which the authors project to continue into the future given population growth. However, it is also important to note that there is not a simple linear relationship between land use and albedo given that urbanization has also increased albedo in some regions. At an extreme, some building projects have specifically aimed to increase surface albedo to deter heat absorption and have been shown to successfully reduce global warming effects (Cotana et al., 2014).

Another driver of surface albedo variability is the presence of natural climate cycles. Among these is a long-term ocean heat cycle which was found to explain a large amount of the global temperature decrease from 1998 to 2013, which was commonly referred to as a global warming hiatus. The 1998-2013 hiatus led many to believe that anthropogenic forcings do not in fact increase global temperature. Kosaka and Xie (2013) show that modifying a GCM to include observed eastern equatorial Pacific sea surface temperatures (SSTs) while estimating all other surface temperatures successfully simulates the hiatus and observed regional patterns. Next, Wyatt and Curry (2014) study the role of the Eurasian Arctic shelf sea ice in explaining the previously-identified stadium wave pattern which appears to cycle around the northern hemisphere every $\sim$50-80 years as indicated through the coordination of a variety of climate indices. Wyatt and Curry (2014) identify variations in the formation of sea ice in the Eurasian Arctic ice shelf as a significant physical mechanism behind the stadium wave, with increased ice formation corresponding to cooling phases of the cycle. Moreover, this finding suggests the stadium wave pattern may be correlated with surface albedo given the large difference in ocean and sea ice albedoes. Similarly, the Atlantic multidecadal oscillation (AMO) is characterized by warm and cool periods in surface temperature (Knight et al., 2006) and could therefore affect albedo due to ice formation.

Finally, SAF amplification of emissions can alter surface albedo. Theoretically, Budyko (1969) and Held and Suarez (1974) give a basis for the SAF and use it to explain potential changes across time in the planet's heat balance including the forming of ice ages and the so-called snowball Earth. Later on the SAF was studied widely (Déry and Brown, 2007; Qu and Hall, 2007, 2014; Thackeray and Fletcher, 2016). Climate scientists are very concerned about the effects of the SAF and give large estimates for it (Thackeray and Fletcher, 2016), while Kaufmann and Pretis (2023) refute previous SAF estimates with an error correction model incorporating temperature, forcing and snow cover data. The SAF estimate from Kaufmann and Pretis (2023) of 3.4% amplification is approximately an order of magnitude smaller than estimates obtained through more complex process-based climate models. Kaufmann and Pretis (2023) attribute the discrepancy to an oversensitivity of snow cover to temperature increases in process-based models and propose several experiments to validate their 3.4% magnitude in future research. Kaufmann and Pretis (2023) estimate that temperature increases due to forcing are amplified by 3.4% because of the SAF. Therefore, it is plausible that as forcing and temperatures have continued to rise, albedo will have decreased across time in an exponential manner because of the feedback loop.

Albedo changes stand to have a large impact on the global energy balance and most, if not all, causes of albedo shifts could be affected by anthropogenic forcing whose largest component is carbon emissions. Regarding the literature on the effects of emissions on the global energy balance, recent studies have highlighted many complexities that affect the carbon pricing decision. First, many researchers have developed and improved IAMs for the pricing of carbon and incorporate insights from climate models to predict future consequences of carbon emissions. Among the methods used are economic forecasts of energy usage for a given carbon tax and physical climate models showing temperature responses to emission targets (Weyant, 2017). Recent advances in this area stem from incorporating detailed regional data into the calculations (Weyant, 2017). More research has been conducted to determine approximate magnitudes of global average temperature increases given specific carbon scenarios and such studies are used by policymakers broadly (Calel and Stainforth, 2017). However, more complex effects of carbon emissions than global average temperature increases have also received attention recently. Among these is polar amplification studied by Brock and Miller (2024) who apply statistical methods to estimate a moist energy balance model (MEBM) with results indicating spatially heterogeneous effects of global average temperature increases when using data from 1850 through 2019. In constructing the MEBM however, Brock and Miller (2024) make a simplifying assumption that incoming energy less albedo at the surface is a constant, which was plausible due to previous evidence indicating little inter-annual variation in surface albedo.

However, physical mechanisms suggesting changing albedo across time raise about the constant albedo assumption of Brock and Miller (2024) and present serious implications for the underpricing of carbon given the presence of similar assumptions in many IAMs. Specifically, if there are large changes or nonstationarity in albedo patterns across time, this could bias the estimates of Brock and Miller (2024) due to an unmodeled component of the error term. Even if our data early in the sample is less reliable, if we find some non-stationary pattern within similar data used by Brock and Miller (2024) that is unmodeled by the temperature and forcing variables, this suggests a problem with their estimates. Specifically, we analyze the most recent versions of the temperature and forcing data sets from Brock and Miller (2024). Although we examine a much more aggregated scale than Brock and Miller (2024) (i.e., global average versus results by latitude), a finding of significant changes in global albedo would still contradict the constant albedo assumption from Brock and Miller (2024).

Regarding the underpricing of carbon emissions, IAMs are commonly used to balance climate and economic growth goals and generate a social cost of carbon emissions for policy use. A serious drawback to this calculation is the lack of an albedo consideration in commonly used models given the evidence in the climate literature that the SAF amplifies effects of carbon emissions (Kaufmann and Pretis, 2023). Many past IAMs failed to include land use albedo effects and thus did not accurately compute costs and benefits associated with afforestation policy tools, where changes in land cover are significant (Jones et al., 2015). However, even when considering albedo effects due to land use, IAMs often still do not include mechanisms for the SAF (Jones et al., 2015). Specifically, in several IAMs used by policymakers (Dynamic Integrated model of Climate and the Economy - DICE; The Climate Framework for Uncertainty, Negotiation and Distribution - FUND; and Policy Analysis for the Greenhouse Effect - PAGE) the climate modules assume a constant climate sensitivity parameter to capture the surface warming associated with doubling the atmospheric CO2 concentration (Calel and Stainforth, 2017). This omits potential nonlinearities such as the SAF given that it is time and state invariant. If the SAF is not accounted for, this can lead to significant underpricing of carbon emissions given that their impacts on climate may be negatively biased from the nonlinearity. Therefore, this study tests whether SAF amplification of carbon emissions is large enough to cause underpricing in IAMs.

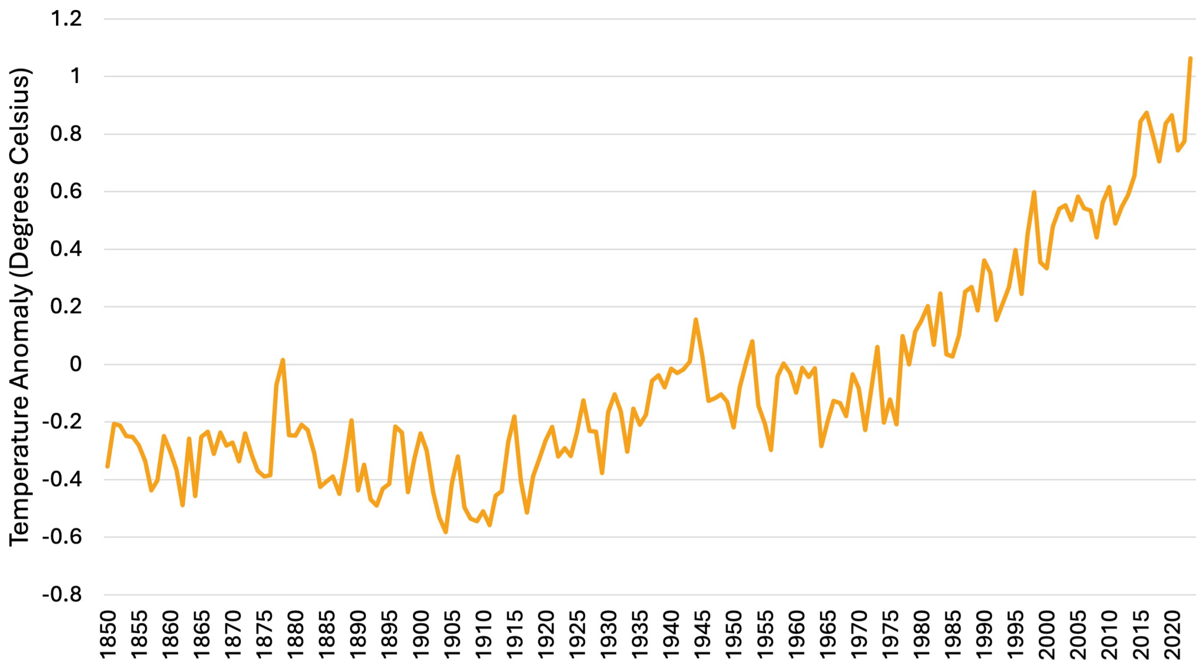

Figure 1: Global average annual temperature anomalies in degrees Celsius relative to 1961-1990 base period from HadCRUT5.

3 Data

3.1 Temperature Anomalies

For temperature anomaly data we use the HadCRUT5 non-infilled annual global average data series[4]. We use HadCRUT 5.0.2.0 non-infilled surface temperature which is a newer version of the HadCRUT 5.0.1.0 non-infilled surface temperature from Brock and Miller (2024), and we use global averages instead of northern hemisphere temperatures used in the previous study. The HadCRUT5 data set combines data from the HadSST sea-surface and CRUTEM land surface temperature data sets. To generate temperature anomalies (in degrees Kelvin), the HadCRUT5 data establishes a base period and displays a deviation from the base period average for each year. Specifically, HadCRUT5 uses 1961-1990 as the base period due to the best sample coverage occurring during this time and the resulting data series is shown in Fig 1. Changes in Kelvin are equivalent to changes in Celsius, so all temperature changes in this study can also be interpreted as degrees Celsius.

It is important to note that temperature measurement methods have evolved significantly over time. Prior to WWII, sea-surface temperatures were largely collected by taking the temperature of sea water extracted using uninsulated buckets from the ocean (Folland and Parker, 1995). After

[4]HadCRUT version 5.0.2.0 “GL” series accessed on 2025-04-19 at https://crudata.uea.ac.uk/cru/data/temperature/

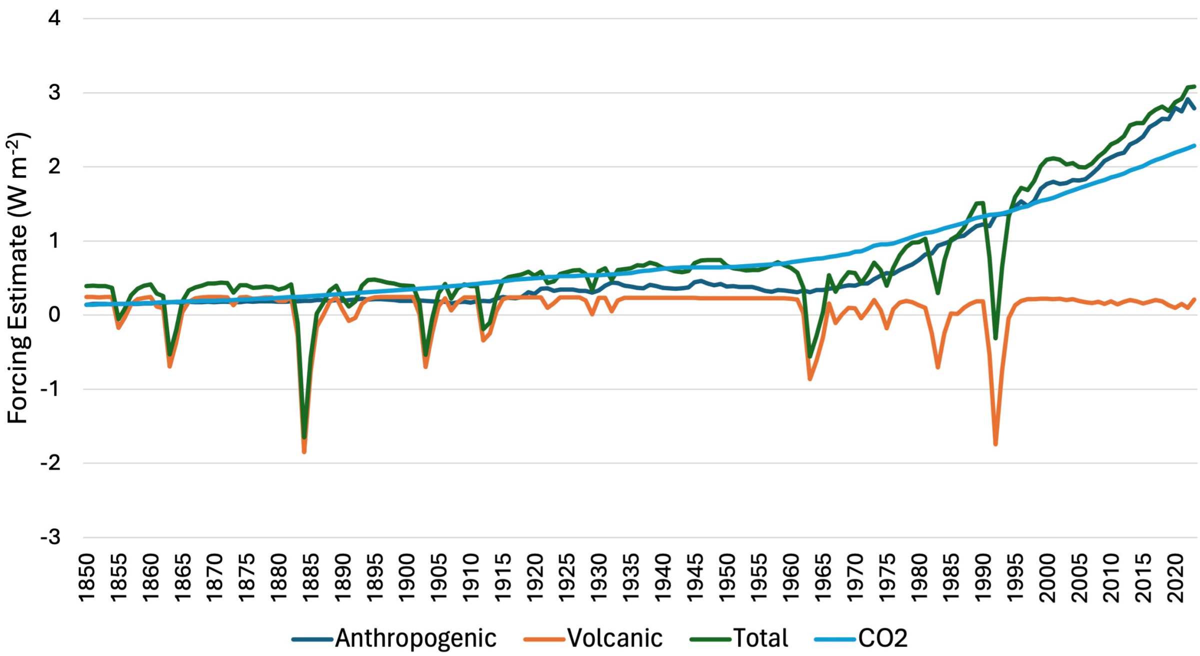

Figure 2: Effective radiative forcing from Forster et al. (2024) shows the effect of each forcing category on the TOA energy balance relative to 1750. Total category includes natural (solar and volcanic) and anthropogenic forcing.

WWII, temperature collection transitioned to more reliable methods including collection methods built directly into the ship (i.e., engine inlets), insulated buckets (Folland and Parker, 1995) and more recently, the use of buoys (Kennedy et al., 2011). The widespread use of uninsulated buckets in sea-surface temperature collection prior to 1940 biased temperatures downward because heat escaped from the water (Folland and Parker, 1995) and for this reason the HadSST team conduct bias corrections based on estimated shares of different collection types combined with physical models of heat transfer from the buckets. To this end, the HadSST data set is reported as a 200-member ensemble by varying the unknown bias correction parameters within reasonable ranges (Kennedy et al., 2019). Regarding uncertainties in the CRUTEM data, the primary change in land surface temperature measurements has been increased station coverage across time (Jones and Moberg, 2003) which has also occurred for sea-surface temperatures (Kennedy et al., 2011). As with HadSST, the HadCRUT5 data set reports a 200-member ensemble to account for systematic bias in measurements, as well as a mean temperature series of the ensemble which is used in this study.

3.2 Radiative Forcings

We use effective radiative forcing (ERF) data generated by Forster et al. (2024)[5] which includes multiple types of anthropogenic forcing and natural forcing from solar and volcanic events. Key ERF series are shown in Fig 2. Forcing data from Forster et al. (2024) updates forcings from Ch 7 of AR6 WGI which were used in Brock and Miller (2024). ERF is radiative forcing that is adjusted to include additional downstream effects of the forcings which are independent of surface temperature effects to maintain separation from climate feedbacks such as the SAF. The Intergovernmental Panel on Climate Change (IPCC) shifted to the use of ERF as its primary forcing measure in AR5 because it includes broader effects of radiative forcings and continued its use in AR6 (Forster et al., 2021). The ERF data is provided in W m^{-2} and can be interpreted as the effect of a forcer on the TOA energy balance since 1750, where positive values indicate energy added to the system. ERF data is generated using simulations in radiative transfer models where the climate is shifted by a forcer (Forster et al., 2021).

3.3 CERES Surface Flux

We use CERES-EBAF edition 4.2.1 monthly global mean computed surface fluxes[6]. According to Kato et al. (2018), the CERES surface flux data is generated from TOA observations by a calibration of computed TOA fluxes to match observed TOA fluxes in which surface, cloud and atmospheric properties are adjusted until computed TOA fluxes match observed CERES TOA fluxes. Then the computed surface fluxes are further modified using both bias and Lagrange multiplier corrections. Finally, the surface flux algorithm is validated by comparing time series of computed surface fluxes to observed surface fluxes collected at a variety of land and ocean surface sites. The computation incorporates cloud and atmospheric data from various sources including MODIS, geostationary satellites, the Goddard Earth Observing System (GEOS) and the Model of Atmospheric Transport and Chemistry (MATCH). Additionally, the CERES team makes changes over time to the data product through modifications to both TOA collection and surface computations. For instance, Kato et al. (2025) was dedicated to addressing issues in the data related to transitions among satellite structures. Early in the sample (March 2000-July 2002) a single pole-to-pole satellite was used, followed by a period where two satellites were in the system, and recently CERES transitioned again to a single satellite (after April 2022). The CERES team adjusted the surface computation in edition 4.2 to account for bias caused by using a single satellite in the presence of strong day/night cloud cycles.

Several of our model specifications include net shortwave flux at the surface from the CERES-

[5] Data version v2024.05.29b accessed on 2025-04-19 from https://zenodo.org/records/11388387

[6] Downloaded on 2025-05-29 from https://ceres.larc.nasa.gov/data/

EBAF product (hereafter referred to as the CERES data). CERES data is available from March 2000 to the present and we aggregate to an annual level using only data from 2001 to 2023. Net shortwave flux at the surface is equivalent to incoming minus outgoing shortwave radiation, which is specifically relevant to albedo because energy from the sun that reaches the surface is in the form of shortwave radiation (Trenberth et al., 2009) and reflected solar radiation remains in shortwave form. CERES measurements contain somewhat limited spatial-temporal resolution (1-degree spatial, monthly data). However, this does not limit us as we use global averages and aggregate to an annual level. Among the strengths of the CERES data are that the system was built specifically for the study of the Earth energy budget and is therefore equipped with broadband sensors (Smith et al., 2011) as opposed to narrowband sensors, which later need to be converted to broadband values for use in energy balance studies (Qu et al., 2015). Moreover, CERES instruments allow for in-flight calibrations which reduce measurement errors.

4 Methods

4.1 Zero-dimensional Energy Balance Model (EBM)

Our model is derived from the zero-dimensional benchmark EBM in Brock and Miller (2024) which is a simple model showing how changes in temperature are a function of incoming energy to the planet, outgoing radiation and forcing. The term zero-dimensional indicates that we do not examine separate data by latitude. The underlying theory behind an EBM is the first law of thermodynamics which says that energy cannot be created or destroyed. An EBM is accounting for where the energy goes in the Earth system and it says that when forcing happens (*i.e.,* an imposed or exogenous change in the energy system – Castle et al. (2024), Brock and Miller (2024)) then temperature has to change (North and Kim, 2017). Although a very simple representation, EBMs are commonly used to investigate specific aspects of the climate and are often used to study the effects of forcing on aggregate temperature. Given the simplicity of EBMs, it is easier to extract macroscopic mechanisms which could broadly affect future research and policy decisions (North and Kim, 2017).

Our baseline EBM is given by

$$C\partial_t T(t) = Q(t) - (A + BT(t)) + E(t) \tag{1}$$

where C is a heat capacity parameter, $T(t)$ and $E(t)$ are surface temperature and forcing at time t, $A + BT(t)$ is outgoing radiation and $Q(t)$ is incoming radiation net albedo at time t. In order to study albedo changes across time, we modify the zero-dimensional EBM from Brock and Miller (2024) by converting a constant Q to $Q(t)$ which varies over time. Given that our forcing data is

Table 1: Symbols, units and descriptions of the main quantities, variables, and parameters of the climate and econometric models from Brock and Miller (2024)

Symbol	**Unit**	**Description**
Energy Balance Models		
$E(t)$	W m^{-2} (Watts/square meter)	Forcing
$T(t)$	K (Kelvin)	Surface temperature
$\partial_t T(t)$	K ∇^{-1} (Kelvin per time increment)	Temperature increase
$C > 0$	W m^{-2} K^{-1} ∇	Heat capacity
$Q(t) > 0$	W m^{-2}	Incoming radiation less albedo (reflection)
$A > 0$	W m^{-2}	Intercept parameter (removed in estimation)
$B > 0$	W m^{-2} K^{-1}	Sensitivity of outgoing long-wave radiation
E_t^*	W m^{-2}	Forcing anomaly
T_t^*	K	Surface temperature anomaly
Q_t^*	W m^{-2}	Incoming radiation less albedo anomaly
State-Space Models		
$y_{t+1} = \Delta T_{t+1}^*$	K ∇^{-1}	Change in surface temperature from year t to $t+1$
$\alpha_1 = -B/C$	∇^{-1}	Coefficient on surface temperature anomaly
$\alpha_2 = 1/C$	W^{-1} m^2 K ∇^{-1}	Coefficient on non-volcanic forcing anomaly
$\alpha_3 = 1/C$	W^{-1} m^2 K ∇^{-1}	Coefficient on volcanic forcing anomaly
$\alpha_4 = 1/C$	W^{-1} m^2 K ∇^{-1}	Coefficient on incoming radiation less albedo anomaly
$E_t^{*(novolc)}$	W m^{-2}	Non-volcanic forcing anomaly
$E_t^{*(volc)}$	W m^{-2}	Volcanic forcing anomaly
G_t^*	K ∇^{-1}	Heat effect of incoming radiation less albedo anomaly
S_t^*	K ∇^{-1}	Heat effect of incoming radiation less albedo anomaly controlling for land use, BC on snow and solar forcings

available at an annual level, we modify the EBM to

$$C(T_{t+1} - T_t) = Q_t - (A + BT_t) + E_t \tag{2}$$

which is now in discrete time with increments of one year.

Next, we represent the EBM in terms of anomalies (changes) relative to an arbitrary base forcing $\bar{E}$, here defined as the 1850-2023 sample average, with forcing anomalies $E_t^* = E_t - \bar{E}$. Moreover, the anomalies relative to base forcing for temperature and incoming radiation less albedo are $T_t^* = T_t - \bar{T}$ and $Q_t^* = Q_t - \bar{Q}$ where $\bar{T}$ and $\bar{Q}$ are sample period averages. To start, the base period equilibrium is subtracted from the EBM and gives

$$\begin{array}{c} C(T_{t+1} - T_t) = Q_t - (A + BT_t) + E_t \\ \underline{-(C(\bar{T} - \bar{T}) = \bar{Q} - (A + B\bar{T}) + \bar{E})} \\ C(T_{t+1}^* - T_t^*) = Q_t^* - BT_t^* + E_t^* \end{array} \tag{3}$$

where the formula under the horizontal bar is the anomaly representation. The explanation for why the second formula in equation (3) is an equality even though it appears the LHS is zero while the RHS is positive (because temperature increases over our sample) is related to the fact that the average T_{t+1} value differs from the average T_t value. For example, if we have $t = 1, 2, 3$, but our model has $T_{t+1} - T_t$, then we can only use $t = 1, 2$, so the average T_{t+1} is $(T_2 + T_3)/2$, whereas the average T_t is $(T_1 + T_2)/2$. The assumption that incoming radiation less albedo is relatively constant from Brock and Miller (2024) can be summarized as the assumption that $Q_t^* = 0$.

The temperature data and forcing data are available from 1850-2024 and 1750-2023, respectively, so our final data sample extends from 1850-2023. We generate first-differenced temperatures and crop the sample, and subsequently demean series to generate anomalies relative to the sample average, with the exception of first-differenced temperatures which are already in anomaly form. Rewriting the EBM in anomaly form allows us to simplify estimation by removing the extra parameter A. According to Brock and Miller (2024), the temperature data shows surface temperature and the forcing data shows the net effect of these forcings on surface energy so in equation (3) Q_t^* is incoming energy less albedo at the surface relative to a base period average $\bar{Q}$.

4.2 State-Space Model (SSM) and Kalman Filter Estimation

In order to estimate the unobserved variable Q_t^* we use a SSM which is a type of model that represents a system as depending on an unobserved variable referred to as the state variable. In the zero-dimensional EBM shown in equation (3), temperature, forcing and incoming radiation are all observable, but incoming radiation less albedo at the surface is hard to measure especially in the pre-satellite era. A SSM consists of the state equation which shows how the state variable evolves over time and the measurement equation which shows how an observed outcome variable (called the observation or measurement variable) changes over time based on the state variable and other observed covariates (Durbin and Koopman, 2012). The measurement variable is the change in temperature anomaly over a year denoted as $y_{t+1} = T^*_{t+1} - T^*_t$, and the vector of observed covariates x_t includes forcing and temperature anomalies (E^*_t, T^*_t). In the coming sections, the notation

$$Q^*_{t|t} = E[Q^*_t|\mathcal{F}_t] \tag{a}$$

$$y_{t+1|t} = E[y_{t+1}|\mathcal{F}_t] \tag{b}$$

$$\omega_{t|t} = E[(Q^*_t - Q^*_{t|t})^2|\mathcal{F}_t] \tag{c}$$

$$\Sigma_{t+1|t} = E[(y_{t+1} - y_{t+1|t})^2|\mathcal{F}_t] \tag{d}$$

will be used, where the first two equations are conditional expectations given the information available at time t, which is defined as the σ-field $\mathcal{F}_t = \sigma((y_s)^t_{s=1}, (x_s)^t_{s=1})$. The conditional forecast error variances of Q^*_t and y_{t+1} are given by the latter two equations. Moreover, we solve for the steady state (i.e., time-invariant) value of the predicted conditional variance $\omega_{t|t}$ where $\omega_{t|t} = \omega_{t+1|t+1} = \omega_{00}$ (for details see Appendix B.3). With this, we also obtain steady state values for $\Sigma_{t+1|t} = \Sigma_{t+2|t+1} = \Sigma$ and the updated conditional variance of the state $\omega_{t|t+1} = \omega_{t+1|t+2} = \omega_{01}$.

Our first SSM specification treats Q^*_t as fully latent and we omit solar, land use and BC on snow forcings so that the unobserved state variable will absorb effects from these variables. This is done for accurate comparisons across fully and partially latent models, which use satellite estimates of Q^*_t that include effects of changing solar radiation, land use and BC on snow. Our state variable is the heat effect of incoming radiation net albedo anomaly $G^*_t = Q^*_t/C$. Solving for ΔT^*_t in equation (3) and adding an error term u_{t+1} gives the measurement equation in equation (4), where $\alpha_1 = -B/C$, $\alpha_2 = 1/C$ and ΔT^*_{t+1} is replaced with y_{t+1} for notational simplicity. All anomalies are relative to 1850-2023 sample average. The SSM given by

$$Measurement: y_{t+1} = \alpha_1 T^*_t + \alpha_2 E^{*(novolc)}_t + \alpha_3 E^{*(volc)}_t + G^*_t + u_{t+1} \tag{4}$$

$$State: G^*_t = G^*_{t-1} + v_t \tag{5}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{6}$$

is estimated using the Kalman filter which consists of predicting the state and measurement variables one step ahead and then updating the predictions using new information from the measurement variable. The unknown parameters $(\alpha_1, \alpha_2, R, H)$ along with the initial condition of the state variable $G^*_{0|1}$ are estimated using Maximum Likelihood Estimation (MLE). Finally, the estimates of G^*_t are smoothed through Kalman smoothing which incorporates all information from the sample by changing estimates moving backwards. Details on estimation for the fully latent specifications are shown in Appendix B.1.

Here we rescale Q^*_t by $1/C$ implicitly in the SSM by not estimating a coefficient on G^*_t in the measurement equation which allows us to estimate separate coefficients α_2 and α_3. This rescaling means that while Q^*_t in equation (3) was in W m^{-2}, G^*_t is in degrees Celsius per year in equations (4) and (5). In other words, changes in the estimated state will be in terms of the corresponding temperature change. Although G^*_t is thought to include anthropogenic effects that trend over time due to increasing human activity (Kaufmann and Pretis, 2023), there are advantages to modeling incoming radiation less albedo as a pure random walk as opposed to including a broken deterministic trend. This traces back to a debate on modeling climate time series (temperature, forcing) as either following a stochastic trend or a broken deterministic trend. Although several studies have found evidence of a broken deterministic trend in both temperature and forcing (Estrada and Perron, 2014), tests from Kaufmann et al. (2013) reject the nulls of broken deterministic trends against the alternatives of stochastic trends in temperature and forcing series. Moreover, the presence of a stochastic trend in temperature is consistent with physical mechanisms of climate change (Kaufmann et al., 2013). Given that albedo is influenced by temperature and forcing, we model it using a stochastic trend.

For the SSM in equations (4-6), E^*_t has been separated into $E^{*(novolc)}_t$ and $E^{*(volc)}_t$ where $E^{*(novolc)}_t$ contains all forcing anomalies except for volcanic, solar, land use and BC on snow and $E^{*(volc)}_t$ denotes volcanic forcing anomalies. Given that the effects of volcanic eruptions are very large but dissipate quickly, unlike other forcings in the data, there is likely measurement error related to the exact magnitude and timing of volcanic forcing. Therefore, although the separation of volcanic and non-volcanic forcing is not justified theoretically by the EBM where all forcings have a common coefficient $1/C$, this separation has empirical support because measurement error appears to cause attenuation bias. Specifically, when we include volcanic forcing with all other forcings (Appendix Fig A1), the coefficient on total forcing gets much closer to zero (0.1 in Appendix Fig A1 vs 0.3 in Fig 4). Moreover, in Appendix Fig A1 a large upward trend appears in the state estimates corresponding closely to the anthropogenic forcing trend seen in Fig 2. Similarly, previous studies have omitted volcanic forcing or separated it from other forcing variables (Brock and Miller, 2024; Kaufmann and Pretis, 2023). Next, we assume G^*_t follows an AR(1) structure to construct equation (5) and assume errors are iid normal and uncorrelated across observation and state equations in equation (6). The normality assumption is necessary to derive the Kalman filter and justify MLE but is not necessary for use of the Kalman filter. If the normality assumption breaks down, the estimation becomes pseudo-MLE (Miller, 2005). Finally, we run a model where

$E_t^{*(novolc)}$ is redefined to contain land use, BC on snow and solar forcings. With the addition of these forcings, the state remains in degrees Celsius per year. However, the state is now interpreted as the total heat effect of incoming radiation less albedo controlling for land use, BC on snow and solar forcings, which we subsequently refer to as S_t^*.

After obtaining estimates for both G_t^* and S_t^*, we modify the SSM to use CERES annual net shortwave radiation anomalies at the surface from 2001-2023 and make the state variable partially latent. The Kalman filter is known to work well with missing data in the measurement variable (Durbin and Koopman, 2012; Li, 2013; Su et al., 2015), in which case the measurement variable can be considered partially latent. However, to the best of our knowledge a partially latent state is rarely considered in the SSM literature, with our search revealing no studies that modify the KF to work with a partially latent state. Therefore, we developed modifications to the traditional KF procedures and give an overview of these changes below, with further estimation details shown in Appendix B.2.

The partially latent SSM is referred to as the CERES specification and is shown in equations (7-9), where Q_t^* = net shortwave radiation anomalies at the surface. Q_t^* is observed from CERES for t=2001-2023 and therefore the state is partially latent. Specifically, the SSM is identical to the model in equations (4-6) for t =1850-2000 with the minor modification that the state now has a coefficient α_4 in the measurement equation to convert it into W m^{-2} as given in the CERES data. Moreover, we exclude solar activity, land use and black carbon on snow and ice in order to match the aggregate incoming energy net reflection given by the CERES data. Finally, to get anomalies when using CERES data, we set base forcing to the 2001-2023 average forcing and subtract the 2001-2023 average from all the variables (temperature, forcing and net shortwave radiation). The modified SSM is given by

$$Measurement : y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + \alpha_4 Q_t^* + u_{t+1} \tag{7}$$

$$State : Q_t^* = Q_{t-1}^* + v_t \tag{8}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{9}$$

$$y_{t+1|t} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + \alpha_4 Q_{t|t}^* \; for\; t = 1850, ..., 2000 \tag{10}$$

$$\ell_{t+1}(\theta) = -\frac{1}{2} ln(\Sigma) - \frac{1}{2}\frac{1}{\Sigma}(y_{t+1} - y_{t+1|t})^2 \; for\; t = 1850, ..., 2000 \tag{11}$$

$$y_{t+1|t} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + \alpha_4 Q_t^* \; for\; t = 2001, \ldots, 2023 \tag{12}$$

$$\ell_{t+1}(\theta) = -\frac{1}{2} ln(R) - \frac{1}{2}\frac{1}{R}(y_{t+1} - y_{t+1|t})^2 \; for\; t = 2001, ..., 2023 \tag{13}$$

where Q_t^* is equal to its past value plus an error term in the state equation. We run the previously used KF equations for the years 1850-2000. However, from 2001 onward instead of using the predicted state $Q_{t|t}^*$ to predict the measurement variable y (equation 10), we plug in the observed Q_t^* (equation 12). Moreover, we replace Σ in the likelihood (equation 11) with R (equation 13) because $\Sigma = \alpha_4^2\omega_{00} + R$ consists of uncertainty from estimating an unobserved state and uncertainty related to the measurement of y. Σ is the steady-state value of the conditional forecast error variance of y_{t+1} where $\Sigma_{t|t-1} = \Sigma_{t+1|t}$. Therefore, when Q_t^* is observed we remove the uncertainty from estimating an unobserved state variable. We also treat H as a fixed parameter which is estimated using a random walk on CERES data (details in Appendix B.2). Finally, for smoothing we do not smooth the observed Q_t^* but do use the information from observed Q_t^* to smooth the unobserved time periods with details also in Appendix B.2.

Regarding the interpretation of estimated state variables, changes in (G_t^*, S_t^*, Q_t^*) only show changes in surface albedo under several assumptions. First, we assume the heat capacity of the planet C is constant across time. This is a common assumption in the integrated assessment model literature (e.g., both the FUND and PAGE models have constant heat capacity parameters, while DICE has a time-varying heat capacity; Calel and Stainforth (2017)). However, C could be changing over time due to variations in ocean heat uptake capacity (e.g., Kosaka and Xie (2013)). If C is in fact varying over time then the term $-\frac{C_t}{C}\Delta T_{t+1}^*$ would appear in our residuals, shown by

$$(C + C_t)\Delta T_{t+1}^* = Q_t^* - BT_t^* + E_t^* \tag{14}$$

$$(1 + C_t/C)\Delta T_{t+1}^* = \frac{1}{C}Q_t^* - \frac{B}{C}T_t^* + \frac{1}{C}E_t^* \tag{15}$$

$$\Delta T_{t+1}^* = \frac{1}{C}Q_t^* - \frac{B}{C}T_t^* + \frac{1}{C}E_t^* - \frac{C_t}{C}\Delta T_{t+1}^* \tag{16}$$

where it would be difficult to identify time-varying C_t because of estimation of time-varying incoming radiation less albedo Q_t^*. Therefore we leave this extension to future research. Next, our state estimates display changes in albedo to the extent that incoming energy from the sun is constant. In the specifications where the state is S_t^* (i.e., heat effect of incoming radiation less albedo controlling for solar, BC on snow and land use forcings), we account for changes in incoming energy by the inclusion of solar forcing in $E_t^{*(novolc)}$. Finally, the SSMs estimate a time-varying intercept with an AR(1) structure $(G_t^*, S_t^*, \alpha_4 Q_t^*)$ which we interpret as incoming radiation less albedo as motivated by the zero-dimensional EBM from Brock and Miller (2024). However, it is possible that the time-varying intercept is capturing other unmodeled components of the data which follow an AR(1) pattern besides albedo (e.g., if C_t follows an AR(1) pattern, then it would be included in our state estimates).

Moreover, in all above SSMs we fix the AR(1) parameter to equal one. In several model specifications (Figs A1-2, A7-8) we estimate an AR(1) parameter γ so the state equation becomes $Q_t^* = \gamma Q_{t-1}^* + v_t$ and we consistently found the estimate $\hat{\gamma}$ to be very close to one with the exception of Fig A9. $\gamma = 1$ does not necessarily imply that the estimated state contains a unit root. The formula

$$Q_{t+1|t+1}^* = (R/\Sigma)^t Q_{0|1}^* + K \sum_{i=0}^{t-1} \left(\frac{R}{R + \alpha_4^2 \omega_{00}}\right)^i (y_{t+1-i} - \delta x_{t-i}) \text{ for t} \geqslant 1 \tag{17}$$

shows that setting the AR(1) coefficient on Q_t^* to one does not restrict the predicted state to be nonstationary where $Q_{t+1|t+1}^*$ is the predicted state at time t+1 given information at time t+1, Σ is steady state value of the forecast error variance, $Q_{0|1}^*$ is the initial value of the state, K is the steady state Kalman gain, and x_t is a vector of covariates consisting of T_t^* and E_t^*. Instead, $Q_{t|t}^*$ is nonstationary iff $(y_{t+1-i} - \delta x_{t-i})$ is nonstationary because $(\frac{R}{R+\alpha_4^2\omega_{00}})^i$ is absolutely summable. This equation is based on Lemma 3.2.2 of Miller (2005) and the derivation is shown in Appendix B.4. Intuitively, this relationship shows that the estimates of Q_t^* will inherit the properties of the residuals of a model which only includes temperature and forcing. The residuals will contain a unit root if the true Q_t^* contains a unit root. Likewise, if the true Q_t^* does not contain a unit root, then the residuals and state estimates will not contain a unit root.

Lastly, we conduct a statistical test of whether $H = 0$ for all fully latent SSMs where H is estimated using MLE. If the estimated H is statistically different from zero, this is evidence that the state variable changes over time. Specifically, our numerical optimization procedure gives p-values for the significance of $\sqrt{H}$ and we use that p-value to test $H = 0$ given that the null hypotheses $H = 0$ and $\sqrt{H} = 0$ are equivalent.

5 Results

Table 2: Parameter estimates from primary partially latent specification in equations 7-9. Standard errors are in parentheses. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Variable	Estimate
α_1	$-0.679\,(0.089)^{***}$
α_2	$0.267\,(0.045)^{***}$
α_3	$0.076\,(0.027)^{**}$
$\sqrt{\alpha_4}$	$0.197\,(0.039)^{***}$
$\sqrt{R}$	$0.093\,(0.006)^{***}$
H	0.105
$Q_{0\|1}^*$	$-1.282\,(2.294)$

The main results of the paper are shown in Fig 3 where Fig 3a and Fig 3b show unsmoothed

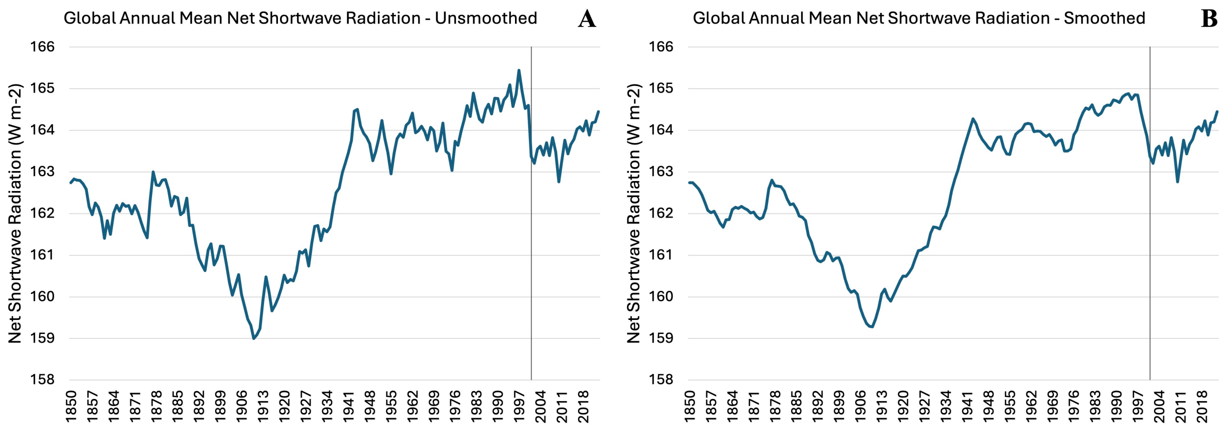

Figure 3: Net shortwave radiation at the surface estimates from primary partially latent SSM in equations 7-9. Estimates in 2001 forward (vertical line) are data from CERES collected via satellite and climate calibration. Prior to 2001, left panel plots the unsmoothed series and right panel plots the smoothed series.

Table 3: Parameter estimates from primary fully latent specifications in equations 4-6. Standard errors are in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Variable	Total Heat Effect (State G_t^*)	Adding Controls (State S_t^*)
α_1	$-0.654\,(0.087)^{***}$	$-0.648\,(0.085)^{***}$
α_2	$0.270\,(0.046)^{***}$	$0.275\,(0.045)^{***}$
α_3	$0.073\,(0.026)^{**}$	$0.070\,(0.026)^{**}$
$\sqrt{R}$	$0.094\,(0.005)^{***}$	$0.094\,(0.005)^{***}$
$\sqrt{H}$	$0.010\,(0.004)^{**}$	$0.010\,(0.004)^{*}$
$G_{0\|1}^* \,/\, S_{0\|1}^*$	$-0.004\,(0.048)$	$-0.006\,(0.046)$

and smoothed estimates. Fig 3 shows the CERES specification which uses CERES net shortwave radiation data from 2001-2023 to back out the net shortwave radiation to 1850. Therefore, from 2001-2023 Fig 3 plots the net shortwave radiation data from CERES. Our primary interpretation uses post-1940 estimates where we have more reliable temperature data. What we can see post-1940 in the unsmoothed estimates is a rough increasing trend in net shortwave radiation from 1940-2023 which corresponds to decreasing reflection from the surface assuming roughly constant incoming solar radiation. However, Fig 3b incorporates more information from the sample than Fig 3a and therefore has more interpretive power. In Fig 3b we again see a rough increasing net shortwave radiation trend from 1940-2023 with a total increase of 0.9 W m^{-2} over this time period. The minimum net shortwave radiation in the post-1940 sample was 162.8 W m^{-2} in 2010 and the maximum was 164.9 W m^{-2} in 1994.

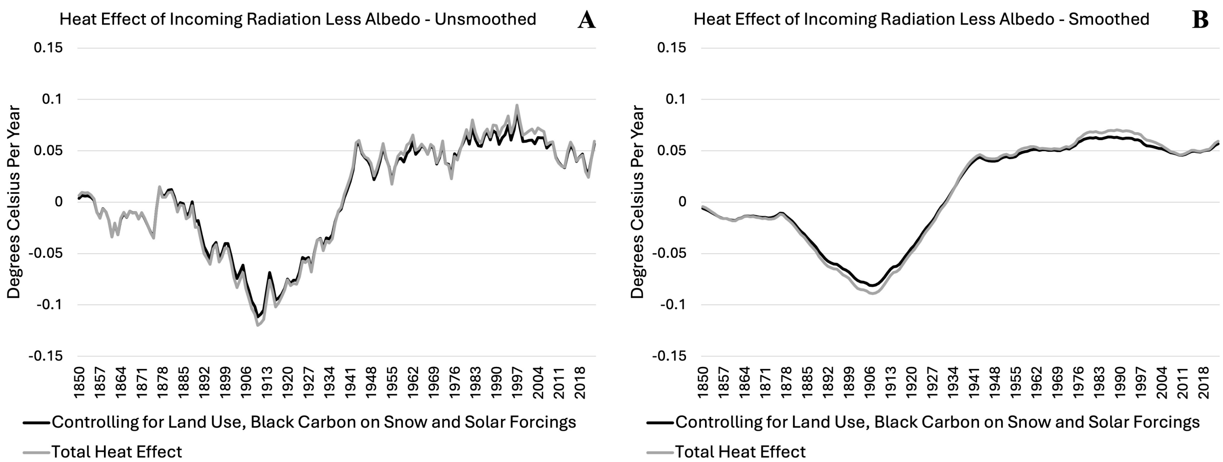

Figure 4: Heat effect of incoming radiation less albedo from primary fully latent SSMs in equations 4-6. Black line is the total heat effect G_t^* and gray line plots S_t^* which includes controls for solar, land use and BC on snow forcings. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

In the unsmoothed primary estimates (Fig 3a) there is a large drop in net shortwave radiation from 2000 when Q_t^* is latent to 2001 when Q_t^* is observed. While it is possible that there was a large drop in net shortwave radiation, this jump is likely a discontinuity which the Kalman Filter takes to be of a reasonable size given the variance of jumps in the earlier estimates (e.g., large decline from 1905 to 1910, large increase from 1935 to 1940). Smoothing does not remove this discontinuity but does spread the decline across a longer timespan. While the partially latent state space model is able to incorporate information from the satellite data in terms of correlation with temperature changes through α_4 and inter-year variation through H, improvements regarding centering the unobserved series on the observed values is a useful area for future research. Results using an estimated H (Appendix Figure A8 and Table A1) have a clearer increasing trend and a smaller decline from 2000-2001 but have noticeable less volatility before 2000 compared to after 2000 which motivates the use of an $\hat{H}$ parameter estimated from CERES data.

Fig 4 shows two model specifications where the difference is included forcings. The gray line shows estimates of G_t^* which include all the same forcings as Fig 3. However, the units of Fig 4 are degrees Celsius per year as opposed to W m^{-2} because the coefficient $1/C$ from the EBM is absorbed by the state variable. The gray line in Fig 4 is our best fully latent comparison model to the CERES specification because we were not able to identify a coefficient on the state variable within the measurement equation. Moreover, there appears to be a mild decreasing reflection trend from 1940-2000 in Fig 4 which is more pronounced in Fig 3. The black line shows estimates for S_t^* which is a robustness specification to compare with past papers that utilize all anthropogenic forcings (e.g., Brock and Miller (2024)) including land use and BC on snow. Moreover, including solar forcings enables us to test the effect of changing solar radiation on our state estimates. The estimates for S_t^* are nearly identical to the estimates for G_t^* which indicates that the variation in our state variable across all models is not due to excluding the forcing components which were controlled for in Brock and Miller (2024). Moreover, controlling for solar forcing does not significantly affect the estimates in Fig 4 which suggests that the large variations in Fig 3 are primarily driven by changes in reflection from the surface as opposed changes in insolation. The unsmoothed estimates in Fig 4a share a similar trend to the smoothed estimates in Fig 4b, except with more inter-annual variation. In Fig 4a from 1940-1995 the change in heat effect from both specifications is ∼0.02 degrees Celsius per year. The heat effect decreases following 1995 and ends at approximately the same level as 1940. At the end of the sample we see a slight increase in Fig 4 estimates which is more clear in Fig 4b. The primary difference between Fig 3 and Fig 4 remains the steeper increasing trend after 2000 in Fig 3, which cannot be explained by the difference in units. Specifically, the 2000-2023 change is ∼0.6 W m^{-2} in Fig 3 and in the grey/black line of Fig 4 it is .002/-.001 degrees Celsius per year. Scaling Fig 3 by $\hat{\alpha_4} = 0.04$ to convert to degrees Celsius per year gives ∼0.02 degrees Celsius per year which is larger than the changes in Fig 4.

Regarding the pre-1940 period, our primary results (Fig 3b) show net shortwave radiation decreasing significantly from 1880-1910 to a low point of 159.2 W m^{-2} and increasing rapidly from 1910-1940. This large dip is consistent across model specifications in Fig 3a and Fig 4a-b. Specifically, the unsmoothed estimates show a similar overall trend with more noise year to year. From 1850-2023 the increase in reflection is 1.7 W m^{-2} which is sizable because CO2 forcings increase by total amount of 2.1 W m^{-2} over this same period. Moreover, the fully latent results show an almost identical trend pre-1940 as Fig 3 with a serious decline in the degrees Celsius per year from 1880-1910 and an increase from 1910-1940. From 1880 to 1910 there was a 0.1 degrees Celsius per year drop and going from 1910 to 1940 there is a 0.15 degrees Celsius per year increase in Fig 4a. In Fig 4b the changes are less severe. From 1880 to 1905 there is approximately a 0.75 degrees Celsius per year decrease and then from 1905 to 1940 about a one degree Celsius per year increase. The appendix section shows that this dip persists across many robustness models including those estimating an AR(1) coefficient on the state, adding an ocean heat cycle as a control and allowing for separate coefficients on major forcing groups. The decrease in net shortwave radiation at the surface from 1880-1910 was 3.4 watts per meter squared in Fig 3b and from 1910-1940 net shortwave radiation increased by 4.3 W m^{-2} to its present (post-2001) range.

Regarding MLE parameter estimates, the underlying parameters for partially and fully latent SSMs are shown in Tables 2-3. From these estimates, the first important thing to note is that the estimates $\hat{\alpha_1}$ are negative and $(\hat{\alpha_2}, \hat{\alpha_3})$ are positive as predicted by the EBM in equation (3) given that both B and C are greater than zero (Brock and Miller, 2024). The estimate for $\hat{\alpha_4}$ was restricted to be positive which is why $\sqrt{\alpha_4}$ is given in Table 2. Next, we can see that the parameter estimates in Table 3 are nearly identical across columns further indicating that solar, land use and BC on snow do not have a large effect on the state estimation. Moreover, the parameters which remain in the same units across partially and fully latent specifications α_1, α_2, α_3, and $\sqrt{R}$ all fall in similar ranges between Tables 2-3. This supports the accuracy of the Kalman filter for estimation of incoming radiation less albedo because even lacking satellite data to discipline estimates, similar results are obtained for the SSM parameters. H and the initial conditions of the state ($Q^*_{0|1}$, $G^*_{0|1}$, and $S^*_{0|1}$) are in different units across Tables 2-3 because the state variable in Table 2 is measured in W m^{-2} while the states in Table 3 are in degrees Celsius per year. Although units are different, the initial conditions of the state share a negative sign indicating an increase in net shortwave radiation at the surface across time.

Moreover, in Table 2 $\hat{\alpha_4} < \hat{\alpha_3} < \hat{\alpha_2}$ which suggests that changes in net shortwave radiation have more measurement error than both non-volcanic and volcanic forcings. Similarly, in both fully latent specifications $\hat{\alpha_3} < \hat{\alpha_2}$. The small magnitude of α_3 compared to α_2 is consistent with the presence of measurement error in volcanic forcing which is dampened by the α_3 coefficient. However, the statistical significance of α_3 at the 1% level shows predictive power of volcanic forcing for temperature changes which justifies its inclusion in the SSM. Similarly, α_4 is statistically significant at the 0.1% level. As noted in Section 3, we examine the statistical significance of H, the variance of the state equation error, to test whether albedo changes significantly over time in the fully latent models. If H is large, this means the state changes a lot between years. If H is small, this means the state changes smoothly. The extreme of $H = 0$ means the state is constant, so a test of $H = 0$ is testing if incoming radiation less albedo is constant. H is statistically significant in Fig 4 at the 5% level for estimates of S^*_t (black line) and at the 1% level for G^*_t (gray line). It is important to note that H is statistically significant in both of the fully latent specifications because this indicates the presence of an unmodeled component in the residuals of the temperature-forcing model and suggests incoming radiation less albedo is not constant across time.

6 Discussion

Table 4: Parameter estimates from typical IAM model in equation (24). Standard errors are in parentheses. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Variable	**Estimate**
α_1	$-0.458\,(0.064)^{***}$
α_2	$0.226\,(0.031)^{***}$
α_3	$0.049\,(0.026)$

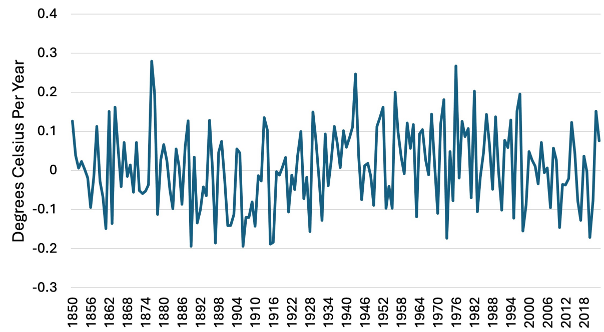

Figure 5: Residuals from typical integrated assessment model in equation (24).

In the IAM literature, the climate sensitivity is how much the earth will heat up due to a doubling of CO2 concentration in the atmosphere. However, this parameter is not modeled as time or state dependent. According to Calel and Stainforth (2017), both FUND and PAGE are based on a one-box climate model which is a zero-dimensional energy balance model with a time-invariant incoming radiation less albedo term of the form

$$\frac{dT}{dt} = \frac{1}{C_{eff}} F - \frac{1}{C_{eff}} \lambda T \tag{18}$$

where F is forcing, T is temperature anomaly and C_{eff} is time-invariant effective heat capacity. λ the feedback parameter which is inversely proportional to the climate sensitivity parameter in equilibrium. Given that IAMs use an equilibrium climate sensitivity, the models back out a single feedback parameter value that is time-invariant. DICE also uses a time-invariant λ term in a two-box model allowing for transfer of heat between the upper and lower levels of the ocean of the form

$$\frac{dT}{dt} = \frac{1}{C_{up}} F_{-} \frac{1}{C_{up}} \lambda T - \frac{1}{C_{up}} \beta (T - T^{LO}) \tag{19}$$

$$\frac{dT^{LO}}{dt} = \frac{1}{C_{deep}} \beta (T - T^{LO}) \tag{20}$$

where T^{LO} is the change in temperature for the deep ocean, C_{up} is the heat capacity of the surface, C_{deep} is the heat capacity of the deep ocean and β is the heat transfer coefficient between the upper and lower ocean.

Given that temperature changes are forecasted as a linear function of past forcing and temperature, if albedo changes as a nonlinear declining function of temperature as predicted by SAF theory, then the declining albedo term will appear in the EBM residuals. More importantly, because forcing causes increases in temperature, additional forcing will cause declines in albedo thereby leading to more rapid heating than predicted by the linear model. IAMs primarily focus on temperature forecasts to evaluate the economic damages of climate change (Calel and Stainforth, 2017) which lead to estimates of the cost of emitting carbon (i.e., carbon prices). If the IAMs underestimate how fast temperature will change in response to forcing, then this will lead to an underestimation of the negative consequences of emitting more carbon and underpricing.

As a simplistic example of the above, consider an IAM that uses a one-box climate approximation given by

$$\frac{dT^*}{dt} = \frac{1}{C}E_t^* - \frac{\lambda}{C}T_t^* \tag{21}$$

with an Euler approximation being

$$\Delta T_{t+1}^* = \frac{1}{C}E_t^* - \frac{\lambda}{C}T_t^* \tag{22}$$

where T_t^* is the temperature anomaly in year t, E_t^* is the forcing anomaly in year t, C is the heat capacity and λ is the time-invariant feedback parameter. Suppose that in reality albedo is a decreasing function of temperature as predicted by SAF theory, leading to a true model of the form

$$\Delta T_{t+1}^* = \frac{1}{C}E_t^* - \frac{\lambda}{C}T_t^* - \frac{Q}{C}\alpha(T_t^*) \tag{23}$$

where $Q\alpha(T_t^*)$ is the incoming radiation less albedo anomaly, Q is constant incoming solar radiation to the surface and $\alpha(T)$ is albedo. Modeling the above climate without including time-varying albedo would result in the term $-\frac{Q}{C}\alpha(T_t^*)$ appearing as an increasing trend in the residuals as temperatures rise. Regarding the implication for carbon prices, surface albedo α decreases due to an increase in forcing from heating effects. If the IAMs set a carbon price as a function of a predicted temperature path in response to forcing such as $P = \mu(T^*|E^*)$ then omitting $\alpha(T_t^*)$ leads to a slower predicted increase in temperatures thereby resulting in underestimation of the cost of carbon emissions.

To this end, partially latent results show an increasing trend in net shortwave radiation from 1940 to 2023. This is in line with climate scientists finding decreased arctic ice extent from 1979 to 2019 (Diebold and Rudebusch, 2023) which would contribute to decreased albedo through the SAF (Kaufmann and Pretis, 2023). However, one limitation to the suggestion of underpricing is that in a typical IAM model run on past temperature and forcing data with the OLS regression

$$\Delta T_{t+1}^* = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + \varepsilon_{t+1} \tag{24}$$

there is no clear increasing trend in the residuals (Fig 5). Compared to our main model estimates (Table 2), the IAM estimates give an α_1 closer to zero (-0.46 vs. -0.68) which indicates the temperature coefficient is absorbing the albedo decline and somewhat accounting for SAF.

Specifically, -$\frac{B}{C}T_t^* - \frac{Q}{C}\alpha(T_t^*)$ is the temperature-related component of the IAM in equation (23). If $\alpha(T_t^*) = -\alpha T_t^*$ (i.e., a decreasing linear function of temperature), the temperature coefficient will absorb the effects of albedo decline and approach zero, becoming $(-\frac{B}{C} + \frac{\alpha Q}{C})T_t^*$. Nonetheless, this model is still prone to underpricing if the SAF mechanism is a nonlinear function of temperature. For example, if the true functional form of albedo is $-\alpha(T_t^*)^j$ with $j > 1$, then temperature projections will be increasingly downward biased as temperature rises. Supporting this, conditional temperature forecasts from IAMs are only a close match to GCM forecasts when the emissions scenarios in question depict smooth changes (Weyant, 2017). Together, this suggests IAMs work well with slow increases in temperature, but the linear approximations degrade with fast changes, as described above with the nonlinear SAF response to large temperature increases.

Moreover, the shorter-term increasing-decreasing reflection trends in our estimates could potentially be explained by endogenous temperature cycles such as the stadium wave and the cycle identified by Kosaka and Xie (2013). The stadium wave pattern studied in Wyatt and Curry (2014) is based on the idea that many previously studied climate cycles coordinate with one another in ~50-80 year cycles. Previous studies found a cycle running around the Northern Hemisphere by longitude (hence the stadium wave analogy where fans in an arena stand up by section). The focus of Wyatt and Curry (2014) is to identify physical mechanisms that could explain the stadium wave pattern. The authors find that the Eurasian Arctic sea ice shelf forming and un-forming explains much of the stadium wave pattern. Given the significance of sea ice in the SAF, it is plausible that cool periods defined by large amounts of sea ice would correspond to increased albedo and hot periods with reduced sea ice would similarly correspond to decreased albedo. Along these lines, Wyatt and Curry (2014) identified 1976-1998 as the warming regime of the stadium wave cycle and 1944-1971 as the cooling regime of the cycle. Looking at the smoothed estimates in Fig 3B, we see that these time periods do in fact show decreased and increased reflectiveness respectively in our estimates. This indicates that the stadium wave pattern seems to significantly correlate with surface albedo and would further suggest that natural cycles feeding into the stadium wave are large explanatory factors in surface albedo thereby contradicting the constant surface albedo assumption of Brock and Miller (2024) and the constant planetary albedo finding of Stephens et al. (2015).

Along these lines, Kosaka and Xie (2013) provide evidence that the 1998-2013 global warming hiatus is caused by cooling in the eastern equatorial pacific ocean. Specifically, they conduct experiments with a GCM where radiative forcings and eastern equatorial Pacific SSTs are observed and allow all other global surface temperatures to be simulated. When restricting the Pacific SSTs, the model closely simulates the global warming hiatus and other important regional climate events. Moreover, given that climate models predict pacific warming in response to the observed 1998-2013 increase in forcings, the authors suggest the pacific cooling is due to a natural cycle. Given the significance of pacific cooling on global surface temperatures, it is plausible that natural ocean heat cycles influence albedo through ice formation. While Kosaka and Xie (2013) do not explicitly mention changes in albedo as an effect of the pacific cooling, the 1998-2013 hiatus corresponds to a period of relatively low net shortwave radiation in Fig 3b. Specifically, from 1998-2013 we find a decreasing pattern in net shortwave radiation with a low point in 2010. However, Kosaka and Xie (2013) also show a decrease in September Arctic sea ice extent over the hiatus period. Together, these factors suggest that pacific cooling may have prevented further sea ice loss than was observed over this period.

Miller and Nam (2020) also study a long-term ocean heat cycle as an explanation of the 1998-2013 global warming hiatus. The authors estimate the long-term cycle referred to as the oceanic multidecadal oscillation (OMO), which shows a peak in $\sim$2023 which is a local minimum of reflectiveness in our estimates. Similarly, Miller and Nam (2020) estimate a trough of the OMO in $\sim$1978 which precedes a period of decreasing reflection in Fig 3b. Therefore, it appears plausible that a long-term ocean heat cycle is a contributing factor to surface albedo. However, as shown in Appendix Fig A5, the estimated OMO cannot explain the entirety of variation in our fully latent estimates of incoming radiation less albedo and is therefore at most a partial contributor, either to changes in surface albedo or as residual unmodeled temperature variation which inflates estimated albedo changes.

Regarding shorter-term climate variability, the El Nino Southern Oscillation (ENSO) is a coupled ocean-atmosphere feedback which causes regional climate variability and whose warm and cool periods are termed El Nino and La Nina respectively. According to Timmermann et al. (2018) the Nino3 SSTA index which is used to determine El Nino events peaked in $\sim$1983, 1988, 1998 and 2016. The El Nino index does not appear strongly correlated with local peaks in net shortwave radiation in Fig 3b. Therefore, ENSO does not seem to significantly affect albedo which could be explained through its diverse regional effects canceling one another in a global aggregate view. Moreover, the findings above solidify a link between global net shortwave radiation and surface temperature variation which is shown through natural surface temperature variation being close to periods of high and low reflection in our data.

Given the above findings of natural temperature variation being correlated with our albedo estimates, we extend our discussion to temperature increases driven by anthropogenic forcings. Along these lines, Diebold and Rudebusch (2023) fit a statistical model to CO2 emissions and arctic sea ice extent data in order to test the relationship between CO2 emissions and arctic sea ice which is a significant contributor to albedo. Diebold and Rudebusch (2023) estimate a linear relationship between CO2 and sea ice and find a strong negative correlation. This supports the correlation between increased temperatures and decreased albedo discussed above because CO2 emissions have been shown to increase surface temperatures through the greenhouse effect (Mitchell, 1989). In addition, the generally declining albedo pattern across our model estimates fits the data on arctic sea ice shrinkage used by Diebold and Rudebusch (2023).

More broadly, the decreasing albedo trend shown in Fig 3b is important because it is not explicitly modeled in much of the carbon pricing literature which is of great policy significance. As mentioned previously, Calel and Stainforth (2017) evaluate the physical assumptions of three widely used IAMs by policymakers. While restrictions vary across models (e.g., DICE allows a time-varying heat capacity C while FUND and PAGE do not), one assumption common to all three IAMs is a constant climate sensitivity parameter to capture surface warming for a doubling of atmospheric CO2 concentration. Moreover, Jones et al. (2015) show that many early IAMs excluded effects of land use albedo, and even upon incorporating this mechanism of climate change, still lack a way to account for changing snow and ice albedo. If climate models fail to account for nonlinearities caused by albedo, then costs and benefits of climate policies will be miscalculated due to an underpricing of carbon emissions. Specifically, the reason the estimated decreasing albedo trend from 1940-2023 in Fig 3b is concerning is that the model in equations (7-9) already controls for components which are trending (temperature and CO2 emissions – see Figs 1-2). Therefore, any trend in our estimates either indicates the presence of an unmodeled forcing or a nonlinear term which amplifies the effects of already-modeled forcings as suggested by the SAF literature.

Regarding the magnitudes of our estimates, when we translate these CERES net shortwave radiation values into albedo as a percentage of incoming solar radiation (Appendix Fig A11) we see the global annual albedo estimates from CERES are approximately 12-13% calculated using all-sky shortwave fluxes at the surface and the surface albedo formula from Stephens et al. (2015). In contrast, Kent and Muttoni (2022) estimate global average albedo to be approximately 15% for the past 30 million years. Their methodology computes the fractions of different surfaces covering the Earth in 10-million-year increments by latitude bands and assumes albedoes for each surface type. Kent and Muttoni (2022) find a largely constant albedo of ∼12% from 120 million years ago to 30 million years ago at the start of the late Cenozoic Ice Age. At this point, their findings indicate albedo rose to 15% which the authors claim is the modern value. The paleoclimate data provides some validation of our estimates in Fig 3a-b as they are within the roughly 12-15% range of Earth albedo values found across the past 120 million years assuming roughly constant insolation.

Regarding the difference between albedo values in Kent and Muttoni (2022) and CERES, it could be said that albedo estimates from CERES data are biased because Kato et al. (2018) report uncertainty in upward and downward SW irradiances of 3 and 4 W m^{-2} which are large compared to decadal mean values of these data (23.3, 187.1). Moreover, the CERES data contains more bias in measurements around the poles (Kato et al., 2018) and was shown to underestimate albedo by 8 percentage points for April-September in the arctic compared to estimates collected at surface sites (Huang et al., 2022). However, the spatial patterns CERES-EBAF total surface irradiances fit with past studies (Loeb et al., 2022; Kato et al., 2025) and the time series of surface irradiances in CERES-EBAF are very close to time series of observations collected at surface sites (Kato et al., 2018) which means that while the CERES data has some bias in albedo, specifically underestimating albedo at the poles, it is reliable in aggregate, especially considering the use of CERES data in evaluating large scale climate models through recent phases of the Coupled Model Intercomparison Project (Huang et al., 2022). Moreover, the bias does not affect our results if it is time-invariant because the trends in albedo would still appear accurately.

Next, our study chose to calculate surface albedo from all-sky fluxes as opposed to clear-sky fluxes which are computed fluxes for a scenario where clouds are omitted (Wild et al., 2019). Although clear-sky fluxes are useful for estimating the influence of clouds on the Earth energy budget (Wild et al., 2019), all-sky fluxes are closer to actual surface fluxes and are more accurate for estimating the SAF (Stephens et al., 2015) meaning all-sky fluxes are best for our application of estimating overall net shortwave radiation for the purposes of concrete policy evaluation.

In Fig 4, controlling for solar forcings does not significantly affect our estimates as seen by the similarity of S_t^* and G_t^* estimates. This suggests that variations in our estimates of the heat effect of incoming radiation less albedo are driven by changes in albedo and not in incoming radiation. Given that our estimates in Fig 3 are similar in shape to the estimates in Fig 4, the suggestion that variations are primarily driven by changes in reflection can be extended to Fig 4. Additionally, Kato et al. (2018) compare CERES-EBAF estimates of downward shortwave radiation at the surface to ocean and land surface observation sites and identify the driver of downward shortwave radiation variability as cloud and atmospheric factors. The downward radiation estimates (Appendix Fig A10a) have a notable minimum in 2010 but are otherwise relatively stable from 2001-2023. This further suggests that changes in the CERES net shortwave radiation data are largely explained by albedo changes.

Next, to contribute to the discussion of whether albedo is in fact constant over time as modeled in much of the carbon pricing literature, we analyze the statistical significance of H. Our results for fully latent specifications in Fig 4 show that there is an unmodeled component in the residuals of the temperature-forcing model which changes significantly over time. Moreover, the statistical significance of H suggests incoming radiation less albedo is not constant, even controlling for more easily measured albedo parameters (i.e., land use and black carbon on snow and ice). This result is confirmed in numerous additional fully latent specifications (Appendix Fig A1-5). This however raises the question of why H is not statistically significant in the partially latent results which do estimate H as part of the MLE procedure (Appendix Figure A8 and Table A1). One possible explanation is that measurement error in the CERES data masks the signal of the state variable. As mentioned previously, Kato et al. (2018) note the uncertainties for global annual average surface irradiances are 3 and 4 W m^{-2} for upward and downward shortwave radiation. These uncertainties are larger than our estimated changes in net shortwave radiation from 1940-2023 and 1850-2023 in Appendix Figure A8, which suggests that the variance of the state is no longer statistically significant in Appendix Figure A8 due to noise in the CERES data obscuring the state signal. Although the statistical significance of the state variance disappears when we incorporate satellite data, the point estimates nonetheless suggest economically significant albedo changes which bias the existing carbon price calculation. From sample minimum to maximum in Appendix Figure A8b, we see an increase of 3.1 W m^{-2} and from 1940-2023 the increase is 1.3 W m^{-2}. Moreover, the increase of 0.9 W m^{-2} from 1940-2023 indicated by our main results in Figure 3b is equivalent to approximately 50% of the increase in CO2 forcing (1.7 W m^{-2}) over the same time period. This is evidence that changes in albedo have large policy significance and must be considered in carbon pricing models, especially because the increasing net shortwave radiation trend indicates a nonlinear component of the climate system which can amplify effects of carbon emissions.

In addition to uncertainties in the CERES data, a second limitation of our data stems from temperature measurements which were more uncertain pre-1940 given different collection practices. For this reason, we restrict our primary interpretation to the post-1940 period. Brock and Miller (2024) use our same temperature data back to 1850, and similar temperature and forcing data has been utilized by the IPCC Sixth Assessment Report (Chen et al., 2021). Therefore, to fully test the constant albedo assumption from Brock and Miller (2024), we include estimates from 1850-1940. Moreover, although the widespread use of uninsulated buckets in sea-surface temperature collection prior to 1940 introduced cold bias into the raw temperatures (Folland and Parker, 1995), the HadSST data set was built using a bias correction to address this. Therefore, while significant uncertainty exists in surface temperature measurements prior to 1940, risks of bias have been mitigated by correction models (Kennedy et al., 2019).

Regarding modeling limitations, it is possible the large magnitude of the 1940-2023 Q_t^* change is due to capturing changes in C_t. Furthermore, the correlations among natural temperature cycles (e.g., from Kosaka and Xie (2013)) and our state estimates could be driven by capturing C_t. Consideration of a time-varying heat capacity is important but beyond the scope of the current research. Therefore, an important future study would be the modeling of a time-varying heat capacity. A second significant future study would be to estimate albedo by latitude possibly using a two-box approximation like Brock and Miller (2024). Moreover, future studies should estimate albedo as a function of temperature similar to Held and Suarez (1974) to estimate the magnitude and functional form of the SAF across 1850-2023. Specifically, it is unknown how to specify SAF within an energy balance model, with previous theoretical work leaving albedo as an undefined decreasing function of temperature (Budyko, 1969) or using a discontinuous function of temperature (Held and Suarez, 1974). Even among highly developed GCMs, there is still disagreement over the strength of the SAF and how to specify the evolution of albedo (Qu and Hall, 2014; Kaufmann and Pretis, 2023). For this reason, future studies should use the albedo trend in Figure 3 to improve SAF specifications.

7 Conclusion

The goal of climate economics is to balance the tradeoff between improved climate and increased economic activity. To this end, IAMs have been developed to measure the costs of carbon emissions and limit economic activity only to the extent that the climate benefits exceed costs of reduced production (Weyant, 2017). Moreover, given recent broad implementation of carbon taxes and tradable pollution permits (Coria and Sterner, 2010; Narassimhan et al., 2018) it is necessary to consider whether the policies put in place are fully measuring effects of carbon emissions on global temperatures. Along these lines, Diebold and Rudebusch (2023) show that large-scale climate models in CMIP5 and CMIP6 underestimate the rate at which we will arrive at an ice-free arctic if carbon emissions continue at current rates. This highlights both a need for research on Arctic climate feedbacks and the potential of even highly complex climate models to underestimate effects from nonlinear feedbacks. IAMs used for cost-benefit analysis of carbon taxes and tradable permits incorporate much simpler climate models than those considered by Diebold and Rudebusch (2023) which allows for increased complexity of economic models (Calel and Stainforth, 2017). However, our findings suggest that the SAF is amplifying the heating effect of carbon emissions, thereby biasing the social cost of carbon from IAMs (Weyant, 2017).

Our results contribute to the carbon pricing discussion because albedo is one of the least documented elements of the Earth's energy balance and we help inform policymakers about this variable. To this end, price is important because it balances the tradeoff between increased production and reduced pollution. We differ from previous studies which examine inter-annual variation in global surface albedo because we use a longer time span from 1850-2023, whereas previous studies have been limited to periods where satellite data is available (Henderson-Sellers and Wilson, 1983; Stephens et al., 2015; Mengyao et al., 2023). This paper leverages both satellite data and time series methods for the estimation of unobserved variables to extend the time frame under consideration. In addition to the carbon pricing literature, our net shortwave radiation estimates could be used by researchers modeling broader forcing-temperature relationships across time into a period before satellites (e.g., Brock and Miller (2024)), which is important for drawing distinctions between natural cycles and anthropogenic effects. In closing, the main finding of our study is that albedo appears to be decreasing from 1940-2023. Given that we control for trends in temperature and carbon emissions, an additional trend in albedo suggests that carbon emissions are being amplified by the SAF and existing models are underpricing carbon emissions by omitting a time-varying albedo term. Again, this occurs due to a decreasing function of temperature (i.e., surface albedo) causing the speed of temperature increases to be underestimated, which leads to underpricing of carbon because temperature is the main input for computing economic damages in IAMs. Therefore, our research highlights the need for continued tracking of albedo over time and the addition of the SAF to contemporary carbon pricing models.[7]

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[7] On a more positive note, changes in climate due to albedo can also influence future economic prospects worldwide due to changes in trade routes and mineral access (Khon et al., 2010; Walsh, 2013; Stephenson and Smith, 2015; Laliberté et al., 2016). Specifically, trade routes such as the Northern Sea Route and the North-West passage which have historically been restricted by extensive ice cover may soon be made open (Khon et al., 2010). Furthermore, frozen locations such as Greenland have desirable minerals which will become more accessible if the area warms (Rosa et al., 2023).

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A ADDITIONAL MODEL SPECIFICATIONS

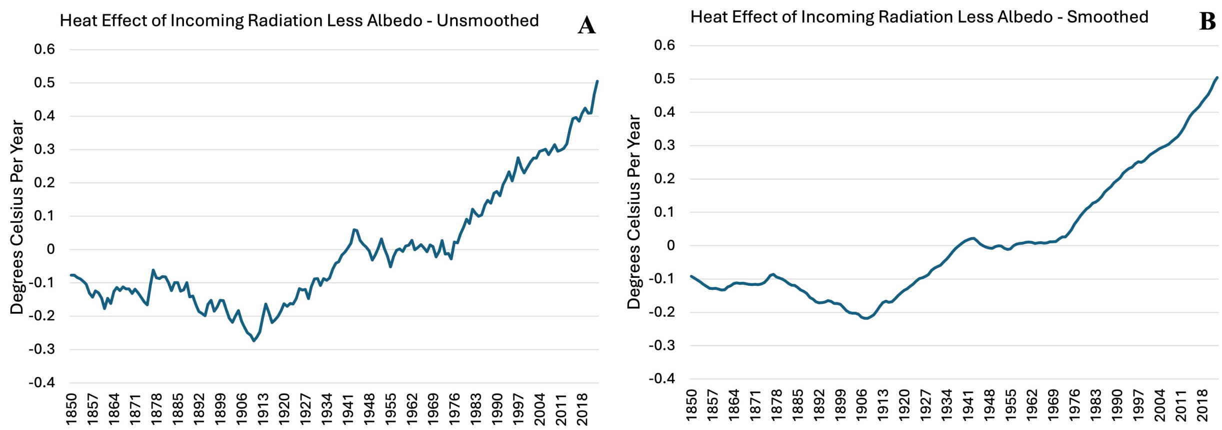

Figure A.1: Heat effect of incoming radiation less albedo from fully latent SSM in equations 24-26. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

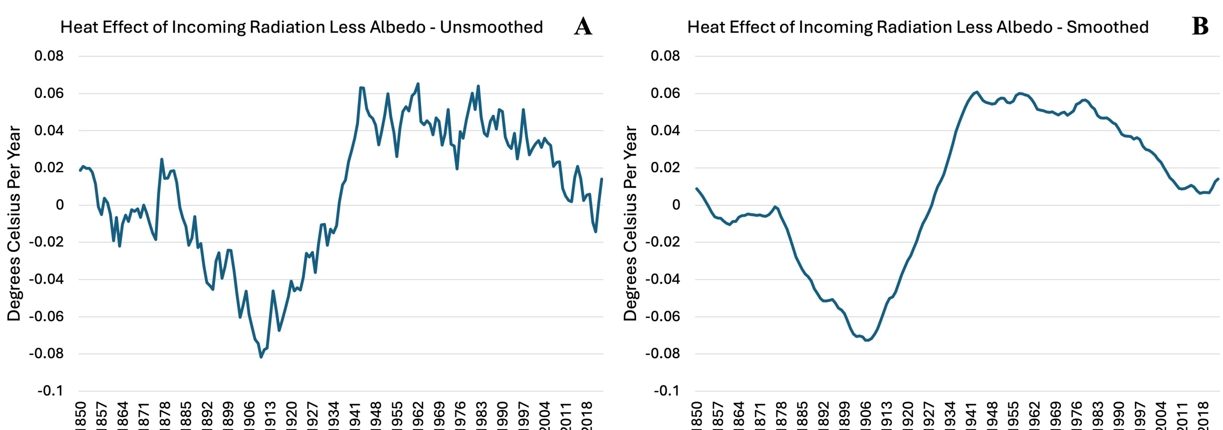

Figure A.2: Heat effect of incoming radiation less albedo from fully latent SSM in equations 24-26 where E_t^* is only anthropogenic forcings. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

The initial specification we attempted was essentially taking a state-space model directly to the EBM shown in equation (3) as follows, where $\alpha_1 = -B/C$, $\alpha_2 = 1/C$, $y_{t+1} = \Delta T^*_{t+1}$, $S^*_t = \frac{1}{C} Q^*_t$

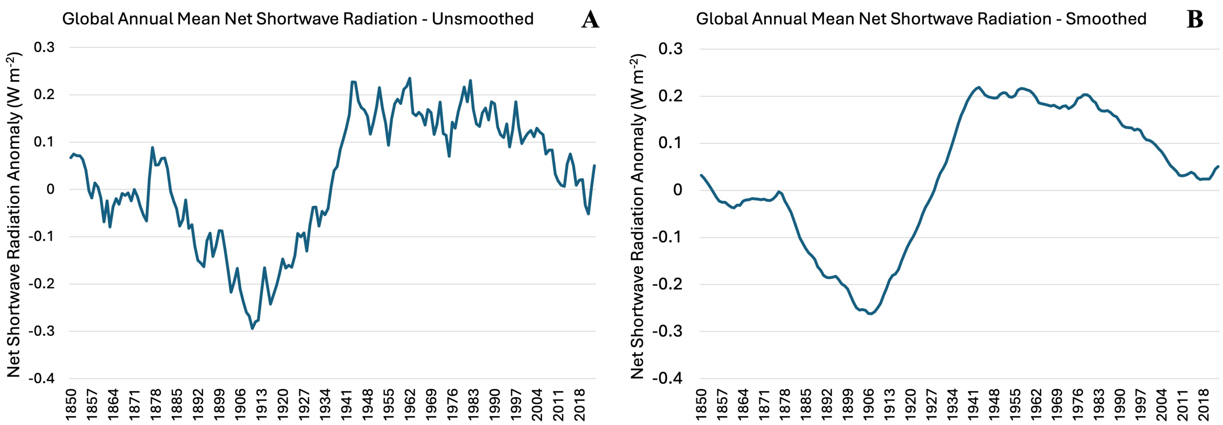

Figure A.3: Net shortwave radiation at the surface estimates from fully latent SSM in equations 27-29. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

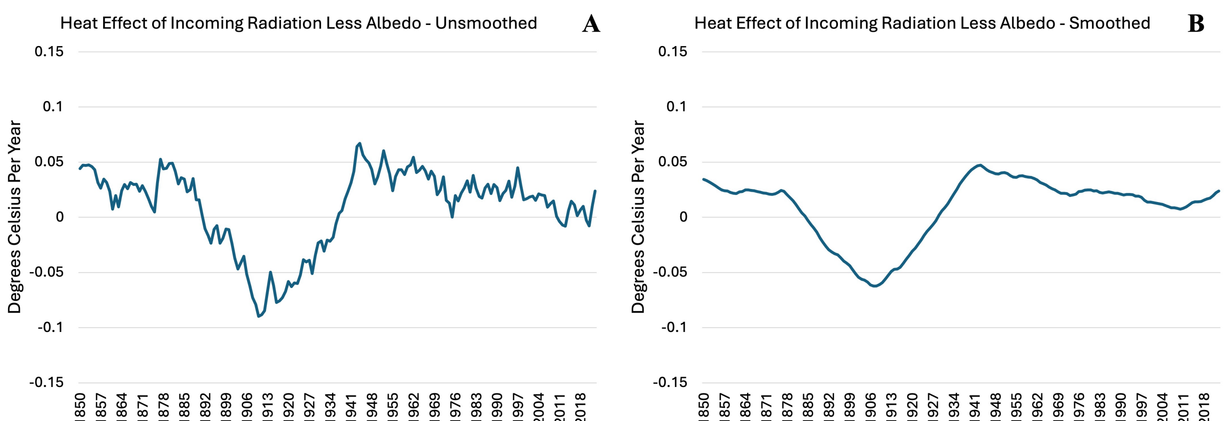

Figure A.4: Heat effect of incoming radiation less albedo from fully latent SSM in equations 30-32. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

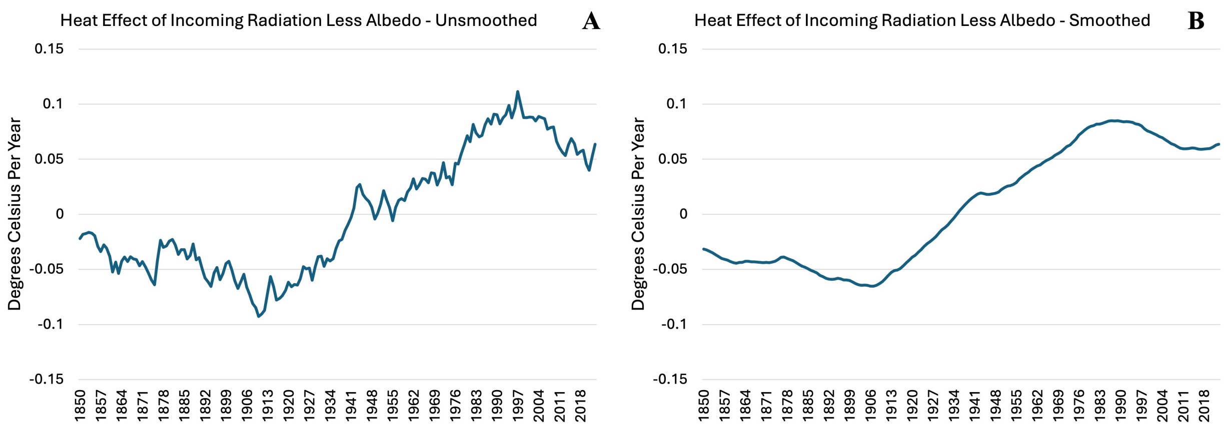

Figure A.5: Heat effect of incoming radiation less albedo from fully latent SSM in equations 33-35. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

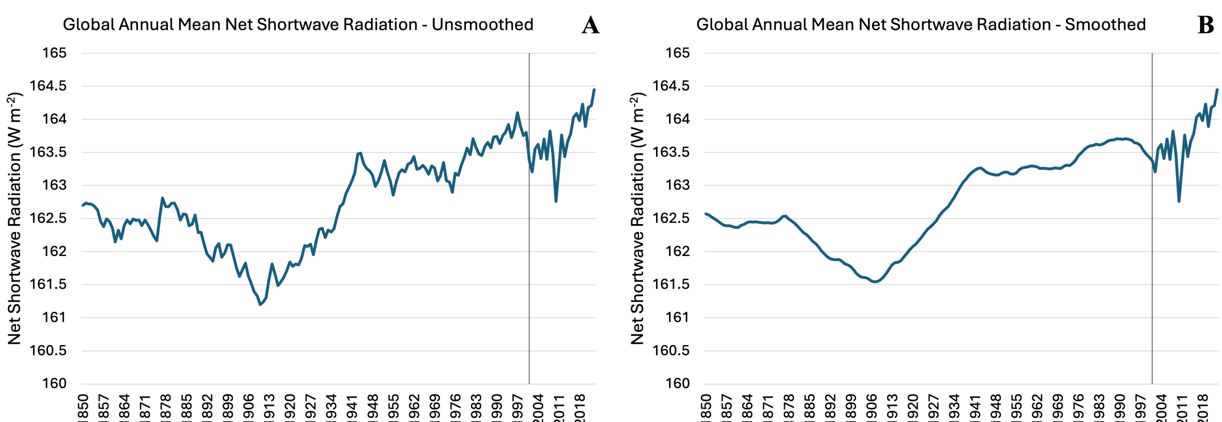

Figure A.6: Net shortwave radiation at the surface estimates from partially latent SSM in equations 7-9 where J is defined as in Appendix B.1 and $\hat{H}$ is replaced with a parameter estimated via MLE. Estimates in 2001 forward (vertical line) are data from CERES collected via satellite and climate calibration. Prior to 2001, left panel plots the unsmoothed series and right panel plots the smoothed series.

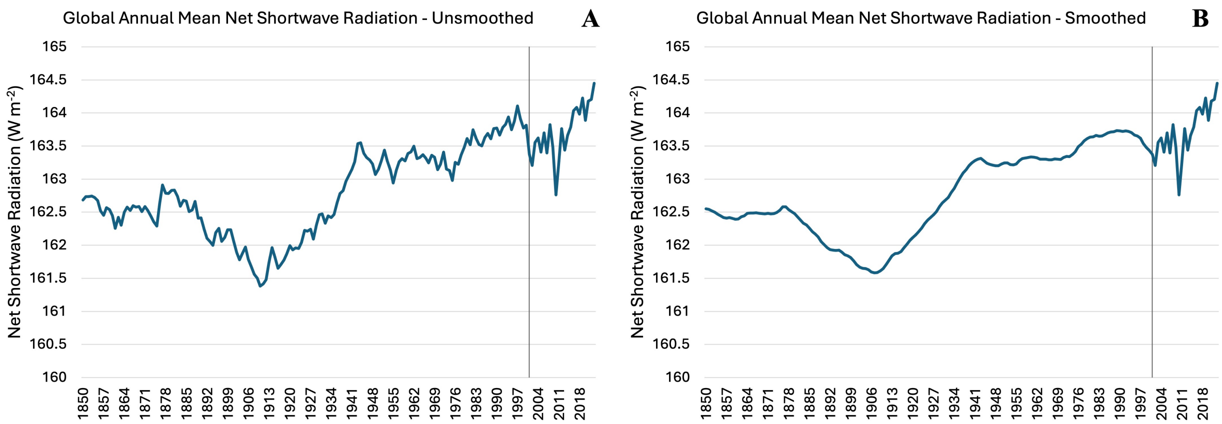

Figure A.7: Net shortwave radiation at the surface estimates almost the same estimates from partially latent SSM in equations 7-9 where J is defined as in Appendix B.1, $\hat{H}$ is replaced with a parameter estimated via MLE and an AR(1) coefficient on the state is estimated. Estimates in 2001 forward (vertical line) are data from CERES collected via satellite and climate calibration. Prior to 2001, left panel plots the unsmoothed series and right panel plots the smoothed series.

Table A.1: Parameter estimates from primary partially latent specification in equations 7-9 replacing $\hat{H}$ with a parameter estimated via MLE. Standard errors are in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Variable	**Estimate**
α_1	$-0.687\,(0.090)^{***}$
α_2	$0.242\,(0.056)^{***}$
α_3	$0.076\,(0.027)^{**}$
α_4	$0.099\,(0.065)$
$\sqrt{R}$	$0.093\,(0.006)^{***}$
$\sqrt{H}$	$0.132\,(0.084)$
$Q^*_{0\|1}$	$-1.136\,(1.123)$

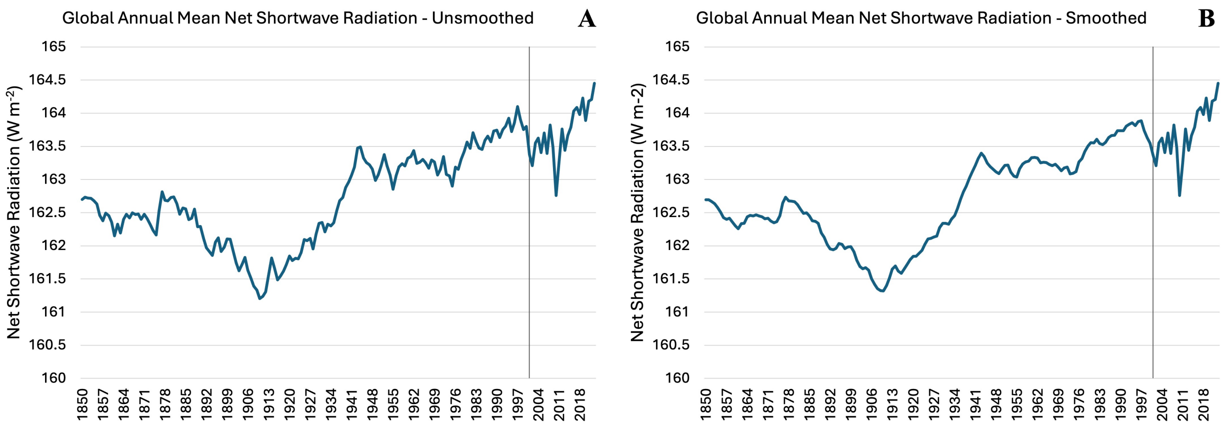

Figure A.8: Net shortwave radiation at the surface estimates from partially latent SSM in equations 7-9 replacing $\hat{H}$ with a parameter estimated via MLE. Estimates in 2001 forward (vertical line) are data from CERES collected via satellite and climate calibration. Prior to 2001, left panel plots the unsmoothed series and right panel plots the smoothed series.

and E_t^* is total forcing as reported by Forster et al. (2024), including volcanic, solar, land use and BC on snow forcings given by

$$Measurement : y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^* + S_t^* + u_{t+1} \tag{25}$$

$$State : S_t^* = \gamma S_{t-1}^* + v_t \tag{26}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{27}$$

where all anomalies are relative to 1850-2023 sample average and results are shown in Appendix Fig A1.

From Appendix Fig A1, we see a strong upward trend in the S_t^* estimates. However, running the SSM above replacing E_t^* with only anthropogenic forcing painted a much different picture with smaller magnitudes of changes in the state (Appendix Fig A2). Given the large sudden changes in volcanic forcing and the exclusion of volcanic forcing from aggregated forcing measures in previous studies (Brock and Miller, 2024; Kaufmann and Pretis, 2023), we separated volcanic forcing from all other forcings in our primary specifications. In Fig 4 we see a striking resemblance to the model using only anthropogenic forcing, indicating that aggregating volcanic forcings with all other forcings significantly alters the state estimate. Moreover, comparing the coefficients α_2 and α_3 within Fig 4 estimates in Table 3 shows that α_3 is much smaller, indicating that the estimated model dampens the noise introduced by volcanic forcing. The separation of the volcanic forcing variable both allows the model to remove extra noise and incorporate information from this forcing

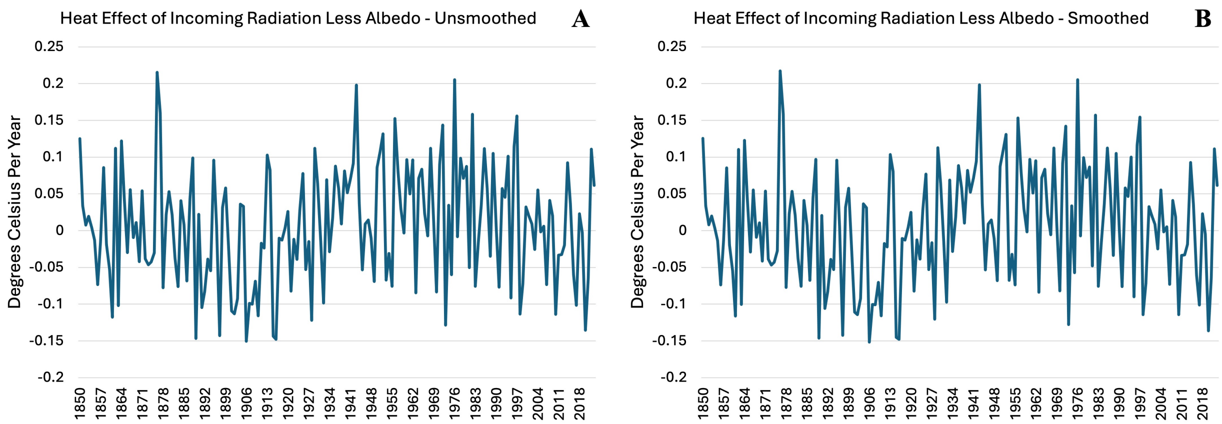

Figure A.9: Heat effect of incoming radiation less albedo from fully latent SSM in equations 4-6 with an estimated AR(1) parameter on the state. Left panel plots the unsmoothed series and right panel plots the smoothed series. Anomalies are relative to 1850-2023 sample average.

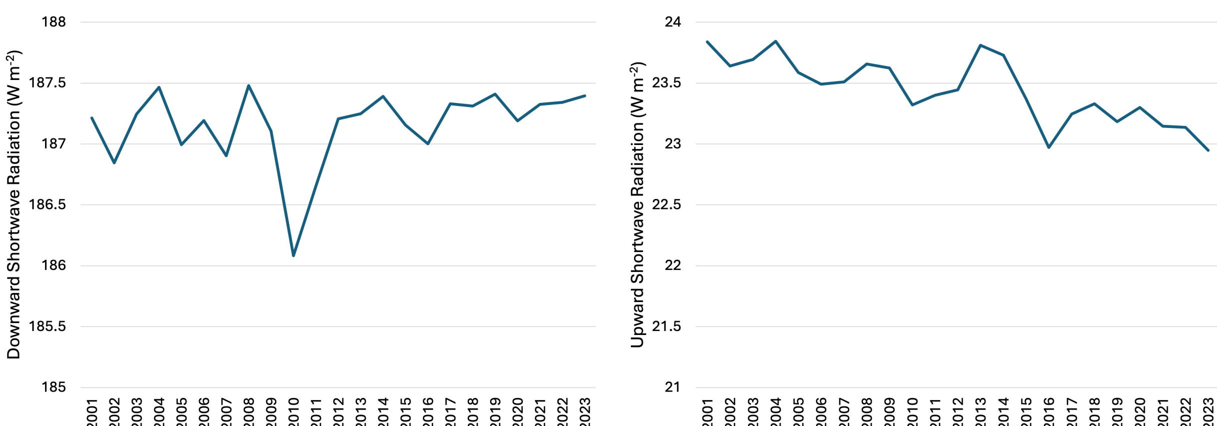

Figure A.10: Downward and upward shortwave radiation at the surface data from CERES collected via satellite and climate calibration.

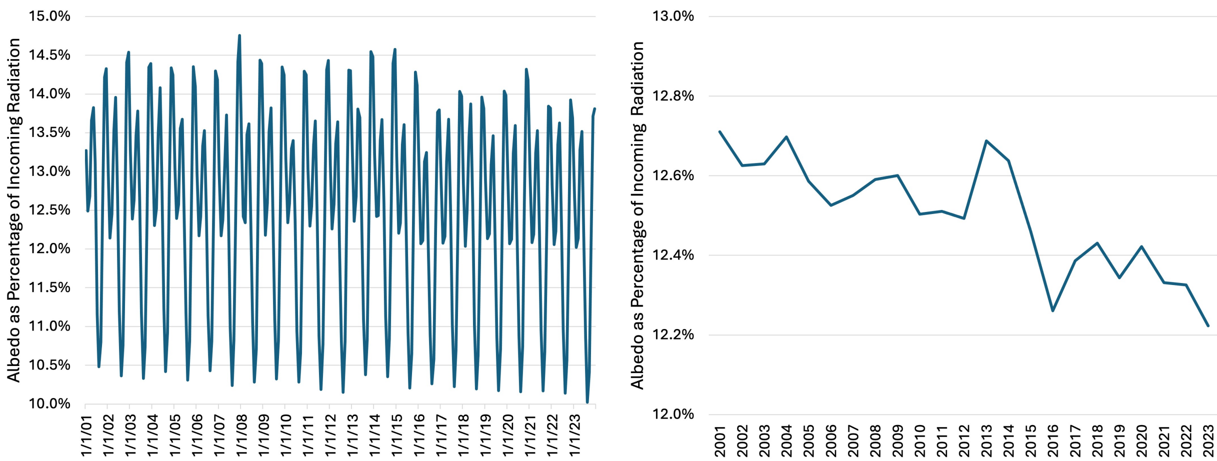

Figure A.11: Surface albedo calculated from CERES data by dividing upward radiation by downward radiation. Left and right panels show monthly and annual values, respectively.

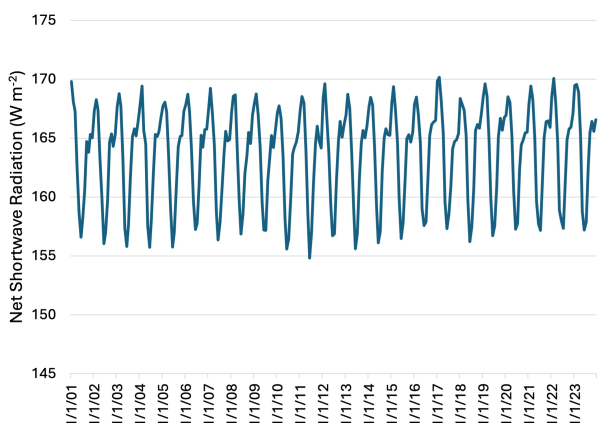

Figure A.12: Monthly net shortwave radiation at the surface data from CERES collected via satellite and climate calibration.

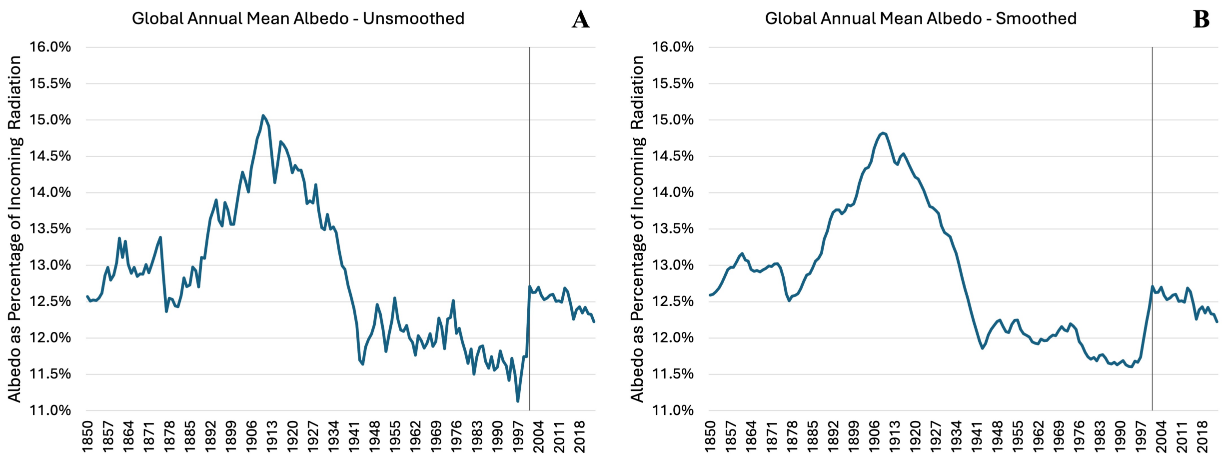

Figure A.13: Surface albedo estimates from partially latent SSM in equations 7-9 modified to use surface albedo as a percentage from CERES and replacing $\hat{H}$ with a parameter estimated via MLE. Estimates in 2001 forward (vertical line) are data from CERES collected via satellite and climate calibration. Prior to 2001, left panel plots the unsmoothed series and right panel plots the smoothed series.

to the extent that it improves model fit and therefore we maintain this structure for the SSMs in the main text. Moreover, we modify the initial models above by fixing the parameter γ to be equal to one. Because $\hat{\gamma}$ estimates given by the above equations are very near one (1.02 and 0.97 respectively), we conclude that it is reasonable to fix γ equal to one without loss of generality.

As robustness, we also modify the above anthropogenic forcing model to become

$$Measurement : y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^* + \alpha_2 Q_t^* + u_{t+1} \tag{28}$$

$$State : Q_t^* = \gamma Q_{t-1}^* + v_t \tag{29}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{30}$$

where a coefficient on the state variable is estimated to convert it to W m^{-2} and force the coefficient to equal the non-volcanic forcing coefficient α_2, with results shown in Appendix Fig A3.

The Q_t^* estimates from the above model retain a similar shape to previous S_t^* estimates without an α_2 coefficient. Moreover, the estimate $\hat{\alpha_2}$ is close to the previous model estimate, thereby indicating that our state estimation is robust to restricting the coefficients on non-volcanic forcing and incoming radiation less albedo to be equal as in the EBM in equation (3). Given this robustness check, for subsequent fully latent models we return to the unrestricted specification where no coefficient is estimated on the state.

Next, because of the large effect that volcanic forcing had on our estimates, we test whether allowing for different coefficients on five major forcing groups significantly alters the results. Specifically, we speculated that aerosol forcing could be causing the large dip from 1880-1940 because it has very large uncertainty in the ERF estimation (Forster et al., 2021). Therefore, we ran the KF on the five-forcing model given by

$$Measurement: y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^{*(volc)} + \alpha_3 E_t^{*(solar)} + \alpha_3 E_t^{*(CO2)} + \alpha_3 E_t^{*(aero)} + \alpha_3 E_t^{*(other)} + S_t^* + u_{t+1} \tag{31}$$

$$State: S_t^* = S_{t-1}^* + v_t \tag{32}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix} \right) \tag{33}$$

where $E_t^{*(volc)}$ is volcanic forcing, $E_t^{*(solar)}$ is solar forcing, $E_t^{*(CO2)}$ is CO2 forcing, $E_t^{*(aero)}$ is aerosol forcing and $E_t^{*(other)}$ is all other forcings and results are shown in Fig A4.

The estimates in Fig A4 show that the general shape of S_t^* is similar when separating the five forcing groups as to when we separate only volcanic forcing in Fig 4. Moreover, the 1880-1940 dip persists with similar magnitudes, thereby indicating that the dip is not due to noise introduced by a specific forcing. A second possible explanation of the dip is that S_t^* could be capturing residual unmodeled temperature variation and therefore does not reflect changes in the heat effect of net shortwave radiation. To test this hypothesis, we incorporate the estimated OMO cycle from Miller and Nam (2020) as a regressor in levels giving

$$Measurement: y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + S_t^* + \alpha_4 O\hat{M}O_t + u_{t+1} \tag{34}$$

$$State: S_t^* = S_{t-1}^* + v_t \tag{35}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix} \right) \tag{36}$$

where the OMO cycle is determined by parameter estimates from Supplementary Table S1 of Miller and Nam (2020) as

$$O\hat{M}O_t = 0.13 sin(13.75(t/T) - 0.21) \tag{37}$$

with $T = 167$ years following the sample length of Miller and Nam (2020).

The results from the OMO specification in Fig A5 show that when controlling for the effects of the OMO, S_t^* still appears to be changing significantly over time. Specifically, in both the smoothed and unsmoothed estimates (Fig A5a and A5b) the dip from 1880-1940 is only partially knocked out, thereby indicating that residual temperature variation does not cause the entirety of our state estimate variation. Additionally, the difference between S_t^* estimates in Fig 4 and Fig A5 could plausibly indicate that surface temperature changes from ocean heat cycles like the OMO alter albedo through melting ice.

Next, when constructing the CERES specification, an initial version of the model was built by estimating the parameter H via MLE and smoothing with J defined as in Appendix B.1 with results shown in Appendix Fig A6. The smoothed estimates with this method appear to have too little year-to-year variation compared to the observed Q_t^* shown in the post-2001 period. Therefore, we use information about year-to-year variation from observed Q_t^* in Figure 3b by replacing J in the smoothing equation with the estimated coefficient $\hat{J}$ from the regression on CERES data given by

$$Q_t^* = JQ_{t+1}^* + \eta_t \tag{38}$$

with further details in Appendix B.2. Estimates using $\hat{J}$ but still estimating H via MLE are shown in Appendix Figure A8, where we see a more plausible variance structure in the pre-2001 years but still less variation than the observed (post-2001) period. We also attempted to modify year-to-year variation in the smoothed estimates by estimating an AR(1) parameter γ in the CERES specification with J specified again as in B.1. However, the estimate $\hat{\gamma}$ remained very close to one (estimate was 0.99) and the resulting Fig A7a appears almost identical to Fig 3a where γ was forced to be one. Moreover, the smoothed version in Fig A7b does not appear to match the observed year-to-year variation any better than Fig A6b. However, Figs A7a and A7b serve as robustness of our state estimates to allowing flexibility in the AR(1) parameter and further validate fixing the AR(1) parameter to one. Compared to the variation shown in the CERES estimates, all above specifications appear to underestimate inter-year variation in Q_t^* thereby motivating the use of a fixed $\hat{H} = var(Q_t^* - Q_{t-1}^*$ estimated with CERES data in the primary specification of Figure 3.

In contrast to the partially latent results (Figs 3, A6-8), fully latent specifications are not robust to estimating an AR(1) parameter. Specifically, we modify the fully latent S_t^* model shown in Fig 4 to estimate a γ parameter, with results shown in Appendix Fig A9. The resulting S_t^* estimates are somewhat similar in shape to Fig 4 but with much more noise and a decreasing net shortwave radiation trend towards the end of the sample. Moreover, the estimated $\hat{\gamma}$ is much smaller than previous models with a value of 0.06. Appendix Fig A7 shows that when satellite data is used, the state estimates are robust to estimating the AR(1) parameter γ and contain much less noise than Appendix Fig A9. The reduction of noise going from Appendix Figs A9 to A7 shows that using Q_t^* data from CERES disciplines the state estimates. Therefore, our main model in Fig 3 incorporates CERES data.

Next, we also attempted to modify the fully latent specification in equations (4-6) by estimating an α_4 coefficient on the state as in

$$Measurement: y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + \alpha_4 Q_t^* + u_{t+1} \tag{39}$$

$$State: Q_t^* = Q_{t-1}^* + v_t \tag{40}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{41}$$

for the purpose of obtaining a direct fully latent comparison model for the CERES specification. However, we were not able to identify both a coefficient on the state and the state variable time series when we have no observed Q_t^* values as indicated by the lack of calculated standard errors in the MLE output. Therefore, our closest fully latent comparison to the CERES specification remains the G_t^* model in equations (4-6).

Next, we present supplementary figures generated from the CERES data. First shown in Appendix Fig A10a and A10b are downward and upward annual shortwave radiation. From these figures, we see a generally decreasing trend in upward annual shortwave radiation, suggesting decreased reflectiveness of the surface. Moreover, the downward shortwave radiation estimates are relatively constant over time with the exception of the 2010 minimum further supporting the idea that net shortwave radiation changes at the surface from 2001-2023 were caused by albedo shifts. Next, we calculate surface albedo as a percentage of incoming energy to the surface which is reflected with the formula

$$\alpha_S = \frac{F_S^{\uparrow}}{F_S^{\downarrow}} \tag{42}$$

from (Stephens et al., 2015), where α_S is surface albedo and $F_S^{\uparrow}$ and $F_S^{\downarrow}$ are upward and downward shortwave radiation at the surface.

In Figs A11a and A11b we show annual and monthly surface albedo. From Fig A11, we see a clear decreasing trend in surface albedo corresponding to the increasing net shortwave radiation trend in Fig 3. Moreover, it appears that the surface albedo decline has accelerated in recent years, which is consistent with acceleration of TOA albedo declines shown by Hansen et al. (2025) starting around 2015. From the monthly estimates, we see large seasonal variation where albedo has a large peak in November/December and a smaller peak in May. Similarly, we observe a large trough in August and a smaller trough in February. Finally, we show monthly net shortwave radiation estimates from CERES in Fig A12. Again, we see much more variation in monthly data compared to annual data, which provides some context for the large changes in net shortwave radiation at the surface observed in Fig 3b. Specifically, the Fig 3b sample minimum of 159.3 W m^{-2} is within the range of net shortwave radiation values observed in Fig A12 which shows that our estimates are within reasonable ranges for the recent climate history.

Using the annual albedo estimates in Fig A11b, we can modify the partially latent model in Appendix Fig A8 to estimate surface albedo as a percentage of incoming radiation. Specifically, North (1975) develops a one-dimensional EBM where incoming radiation less albedo is modeled as $QS(x)a(x,x_s)$ where x is the sine of latitude, x_s is the sine of the latitude of the ice or snow line, Q is the solar constant divided by 4, $S(x)$ is the mean annual meridional distribution of solar radiation which is normalized so the integral from 0 to 1 is unity and $a(x,x_s)$ is one minus the albedo of the earth-atmosphere system. Moreover, North represents $a(x,x_s)$ as

$$a(x,x_s) = \begin{cases} b_0 & x > x_s \\ a_0 + a_2 P_2(x) & x < x_s \end{cases} \tag{43}$$

where b_0, a_0, and a_2 are fixed parameters based on past studies, b_0 is the absorption coefficient over ice 50% covered with clouds, and $a_2P_2(x)$ allows albedo to vary by latitude for the purpose of capturing effects of zenith angle and mean annual cloud distribution. Aggregating over latitudes with an integral from 0 to 1 gives a simplified term Qa where a is one minus the global average albedo of the earth-atmosphere system (i.e., planetary co-albedo). To use this term in practice, we control for inter-annual changes in incoming solar radiation by controlling for solar forcing. Moreover, while a is one minus planetary co-albedo, our paper aims to estimate surface albedo, which can be done through a modified SSM under the assumption of constant cloud and atmospheric factors between years. Therefore, we modify the partially latent SSM to use the annual surface albedo calculated with CERES data (right panel of Fig A.11) instead of annual net shortwave radiation at the surface. First, the EBM in equation (2) becomes

$$C(T_{t+1} - T_t) = Q(1-\delta_t) - (A + BT_t) + E_t \tag{44}$$

where Q is one quarter the solar constant times the fraction of solar energy that reaches the surface and δ_t is surface albedo. Next the anomaly form is given by

$$\begin{array}{r} C(T_{t+1} - T_t) = Q(1-\delta_t) - (A + BT_t) + E_t \\ \underline{-(C(\bar{T} - \bar{T}) = Q(1-\bar{\delta}) - (A + B\bar{T}) + \bar{E})} \\ C(T^*_{t+1} - T^*_t) = -Q\delta^*_t - BT^*_t + E^*_t \end{array} \tag{45}$$

where δ^*_t is the surface albedo anomaly relative to the base period. Relative to the original partially latent SSM, the only substantial difference is which type of CERES data we input (i.e., albedo instead of net shortwave radiation). Otherwise, estimation can continue as before with state δ^*_t and the coefficient $\alpha_4 = -\frac{Q}{C}$. Additionally, $E_t^{*(novolc)}$ now includes solar forcing to control for inter-annual variation in insolation. Results are shown in Appendix Fig A13. A similar declining reflection trend post-2001 is observed as in Appendix Fig A8, but there is no overall decreasing albedo trend from 1940-2023. The large increasing-decreasing reflection trend from 1880-1940 is present and the maximum value of approximately 15% albedo is within the range of 2001-2023 monthly albedo values.

B DETAILS ON EQUATIONS FOR STATE SPACE MODELS

B.1 Kalman Filter Estimation of Fully Latent Models

Our state variable is the heat effect of incoming radiation net albedo anomaly $G_t^* = Q_t^*/C$ and our measurement variable is the change in temperature anomaly over a year $\Delta T_{t+1}^* = T_{t+1}^* - T_t^*$, which is replaced with y_{t+1} for notational simplicity. Note that estimation steps are equivalent for the SSM with the state S_t^*, except with the addition of solar, land use and BC on snow forcings to $E_t^{*(novolc)}$. The resulting SSM is

$$Measurement : y_{t+1} = \alpha_1 T_t^* + \alpha_2 E_t^{*(novolc)} + \alpha_3 E_t^{*(volc)} + G_t^* + u_{t+1} \tag{46}$$

$$State : G_t^* = G_{t-1}^* + v_t \tag{47}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix} \right) \tag{48}$$

where $y_{t+1} = \Delta T_{t+1}^*$, G_t^* = total heat effect of incoming radiation less albedo in degrees Celsius per year, $E_t^{*(novolc)}$ and $E_t^{*(volc)}$are non-volcanic and volcanic forcing respectively and anomalies are relative to the 1850-2023 sample average.

The SSM given in equations (45-47) is estimated in five steps:

1) Initialization - set initial parameter guesses and define steady state variances

2) Prediction - use information at time t to predict state and measurement one time period ahead

3) Updating - use new information from measurement at time $t+1$ to update the state

4) Likelihood and MLE estimation - collect the log-likelihoods from $t = 1...n$ and maximize the sum of log-likelihoods to get parameter estimates

5) Smoothing - use the information across the whole sample to re-estimate the state vector working from the end of the sample to the beginning

The estimation starts with initialization then repeats steps 2 and 3 a total of n times, where n is the number of time periods in the sample. The repetition of steps 2 and 3 is the Kalman filter which is embedded within MLE in step 4. Lastly, smoothing is conducted to give final state estimates with details on each step below.

1) Initialization
The MLE procedure is initialized by setting the log-likelihood to zero and denoting a vector $\theta_0 = (\alpha_1 = -1/2,\ \alpha_2 = 1/2,\ \alpha_3 = 1/2,\ R = 0.1,\ H = 0.1,\ G^*_{0|1} = 0)$ of initial parameter guesses.

The Kalman filter uses the variables

$$\omega_{t|t} = E[(Q^*_t - Q^*_{t|t})(Q^*_t - Q^*_{t|t})'|\mathcal{F}_t] \tag{e}$$

$$\Sigma_{t+1|t} = E[(y_{t+1} - y_{t+1|t})(y_{t+1} - y_{t+1|t})'|\mathcal{F}_t] \tag{f}$$

$$K_{t+1|t} = \frac{cov(Q^*_t, y_{t+1})}{var(y_{t+1}|\mathcal{F}_t)} \tag{g}$$

$$\omega_{t|t+1} = E[(Q^*_t - Q^*_{t|t+1})(Q^*_t - Q^*_{t|t+1})'|\mathcal{F}_{t+1}] \tag{h}$$

where $\omega_{t|t}$ and $\Sigma_{t+1|t}$ are the conditional forecast error variances of Q^*_t and y_{t+1} given information at time t. $K_{t+1|t}$ is the Kalman gain which shows the weight given to new information obtained by the measurement at time $t+1$. Finally, $\omega_{t|t+1}$ is the updated conditional forecast error variance of Q^*_t using information available at time $t+1$.

We solve for the steady state predicted conditional variance of the state ω_{00} where $\omega_{00} = \omega_{t|t} = \omega_{t+1|t+1}$. ω_{00} can be used to find steady state values for $(\Sigma_{t+1|t}, K_{t+1|t}, \omega_{t|t+1})$ denoted by (Σ, K, ω_{01}) which gives

$$\omega_{00} = \frac{H}{2}(1 + \sqrt{1 + 4\tau})\ where\ \tau = R/H \tag{49}$$

$$\Sigma = \omega_{00} + R \tag{50}$$

$$K = \omega_{00}/\Sigma \tag{51}$$

$$\omega_{01} = \omega_{00} - \frac{\omega_{00}^2}{\Sigma} \tag{52}$$

following the derivation of ω_{00} in section B.3.
2) Prediction
$G^*_{t|t}$ given by

$$G^*_{t|t} = G^*_{t-1|t} \tag{53}$$

denotes the predicted state at time t given information at time t. Similarly, $y_{t+1|t}$ from

$$y_{t+1|t} = \alpha_1 T^*_t + \alpha_2 E^{*(novolc)}_t + \alpha_3 E^{*(volc)}_t + G^*_{t|t} \tag{54}$$

is the predicted measurement variable at time $t+1$ given information at time t. $G^*_{t|t+1}$ denotes the updated state at time t given information at time $t+1$.

3) Updating

$$G^*_{t|t+1} = G^*_{t|t} + K(y_{t+1} - y_{t+1|t}) \tag{55}$$

4) Likelihood and MLE estimation

The log-likelihood for each time period $t = 1, ..., n$ is given by

$$\ell_{t+1}(\theta) = -\frac{1}{2}ln(\Sigma) - \frac{1}{2}\frac{1}{\Sigma}(y_{t+1} - y_{t+1|t})^2 \tag{56}$$

whereby the sum of the log-likelihoods $\mathcal{L}(\theta) = \sum_{t=1}^{n} \ell_{t+1}(\theta)$ is maximized using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm giving the parameter estimates $\hat{\theta}$.

5) Smoothing

The smoothed state estimate at time t is denoted as $G^*_{t|n+1}$ and is obtained with

$$G^*_{t|n+1} = G^*_{t|t+1} + J(G^*_{t+1|n+1} - G^*_{t+1|t+1}) \tag{57}$$

working from $t = n$ to $t = 1$ where $G^*_{n|n+1}$ is the updated state at time n obtained in step 3 above and $J = \frac{\omega_{01}}{\omega_{00}}$.

B.2 Kalman Filter Estimation of Partially Latent Models

The procedure for estimating the state using CERES data is

$$Measurement : y_{t+1} = \alpha_1 T^*_t + \alpha_2 E^{*(novolc)}_t + \alpha_3 E^{*(volc)}_t + \alpha_4 Q^*_t + u_{t+1} \tag{58}$$

$$State : Q^*_t = Q^*_{t-1} + v_t \tag{59}$$

$$\begin{bmatrix} u_{t+1} \\ v_t \end{bmatrix} \sim iid\, N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & H \end{bmatrix}\right) \tag{60}$$

which is similar to section B.1 but treats the state variable as partially latent, where Q^*_t = net shortwave radiation anomalies in W m^{-2} and all anomalies are relative to the 2001-2023 sample average. The primary difference in the SSM is that the state variable Q^*_t is estimated with a coefficient in the measurement equation to convert it from degrees Celsius per year to W m^{-2} which are the units of the CERES data. Moreover, we do not estimate a state variable for $t = 2001, .., 2023$ and instead set Q^*_t to be observed net shortwave radiation from CERES. Details of estimation are given below.

The SSM given in equations (57-59) is estimated using the Kalman filter combined with MLE when the state is unobserved (1850-2000) and only MLE when the state is observed (2001-2023).

The estimation begins with initialization, where the initial log-likelihood is set to zero and the vector of initial parameter guesses is $\theta_0 = (\alpha_1 = -1/2,\ \alpha_2 = 1/2,\ \alpha_3 = 1/2,\ \alpha_4 = 1/2,\ R = 0.1,\ Q^*_{0|1} = 0)$. H is estimated in a preliminary step and treated as a fixed parameter $\hat{H}$ within the MLE procedure. Specifically, we estimate $\hat{H} = var(Q^*_t - Q^*_{t-1})$ where Q^*_t is observed CERES satellite data from 2001-2023. Moreover, the steady state predicted conditional variance of the state is denoted by ω_{00} and can be used to give steady state conditional variances for the measurement variable Σ and updated state ω_{01}, as well as the Kalman gain K given by

$$\omega_{00} = \frac{H}{2}(1 + \sqrt{1 + 4\tau})\ where\ \tau = \frac{R}{H\alpha_4^2} \tag{61}$$

$$\Sigma = \alpha_4^2\omega_{00} + R \tag{62}$$

$$K = \frac{\alpha_4\omega_{00}}{\Sigma} \tag{63}$$

$$\omega_{01} = \omega_{00} - \frac{\omega_{00}^2\alpha_4^2}{\Sigma} \tag{64}$$

following the derivation of ω_{00} in section B.3.

For $t = 1850, ..., 2000$ prediction and updating steps are repeated using

$$Q^*_{t|t} = Q^*_{t-1|t} \tag{65}$$

$$y_{t+1|t} = \alpha_1 T^*_t + \alpha_2 E^{*(novolc)}_t + \alpha_3 E^{*(volc)}_t + \alpha_4 Q^*_{t|t} \tag{66}$$

$$Q^*_{t|t+1} = Q^*_{t|t} + K(y_{t+1} - y_{t+1|t}) \tag{67}$$

and log-likelihoods are calculated by

$$\ell_{t+1}(\theta) = -\frac{1}{2}ln(\Sigma) - \frac{1}{2}\frac{1}{\Sigma}(y_{t+1} - y_{t+1|t})^2 \tag{68}$$

for $t = 1850, ..., 2000$. For the time periods $t = 2001, ..., 2023$, the Kalman filter is skipped and instead we use

$$y_{t+1|t} = \alpha_1 T^*_t + \alpha_2 E^{*(novolc)}_t + \alpha_3 E^{*(volc)}_t + \alpha_4 Q^*_t \tag{69}$$

$$\ell_{t+1}(\theta) = -\frac{1}{2}ln(R) - \frac{1}{2}\frac{1}{R}(y_{t+1} - y_{t+1|t})^2 \tag{70}$$

to generate the log-likelihood. The reason R replaces Σ is that $\Sigma = \alpha_4^2\omega_{00} + R$ includes uncertainty from estimating an unobserved state ($\alpha_4^2\omega_{00}$) which does not apply since the state is observed.

The sum of the log-likelihoods $\mathcal{L}(\theta) = \sum_{t=1850}^{2023} \ell_{t+1}(\theta)$ is maximized using the BFGS algorithm giving parameter estimates $\hat{\theta}$ which are common to both periods when the state is observed and unobserved.

Regarding smoothing, the observed state Q_t^* for $t = 2001, ..., 2023$ is not smoothed. However, the state estimate is smoothed for prior periods by first setting $Q^*_{2001|2002} = Q^*_{2001}$ where Q^*_{2001} is observed net shortwave radiation in the year 2001. Then, we use

$$Q^*_{t|2002} = Q^*_{t|t+1} + \hat{J}(Q^*_{t+1|2002} - Q^*_{t+1|t+1}) \tag{71}$$

working backwards from $t = 2000, ..., 1850$.

When the state was completely unobserved in section B.1, we had

$$J = \frac{\omega_{01}}{\omega_{00}} = \frac{cov(Q_t^*, Q^*_{t+1}|\mathcal{F}_{t+1})}{var(Q^*_{t+1}|\mathcal{F}_{t+1})} \tag{72}$$

where $\mathcal{F}_{t+1}$ is the information set at time $t+1$. The ratio of conditional covariance over conditional variance is equivalent to the OLS regression coefficient given in

$$Q_t^* = JQ^*_{t+1} + \eta_t \tag{73}$$

and therefore we estimate $\hat{J}$ from the OLS regression on observed CERES data.

B.3 Derivation of Steady State Variances

The steady state variance ω_{00} is the same in both fully and partially latent models discussed in sections B.1 and B.2, and is given by

$$\omega_{00} = \frac{H}{2}(1 + \sqrt{1 + 4\tau}) \text{ where } \tau = \frac{R}{H\alpha_4^2} \tag{74}$$

with $\alpha_4 = 1$ in the fully latent specifications. The derivation is as follows:

First, the prediction and updating equations for conditional variance of the state $\omega_{t|t+1} = var(G_t^*|\mathcal{F}_{t+1})$ or $\omega_{t|t+1} = var(Q_t^*|\mathcal{F}_{t+1})$ are given by

$$\omega_{t+1|t+1} = \omega_{t|t+1} + H \tag{75}$$

$$\omega_{t|t+1} = \omega_{t|t} - \frac{\omega^2_{t|t}\alpha_4^2}{\Sigma_{t+1|t}} \tag{76}$$

and the conditional variance of the measurement variable is given by

$$\Sigma_{t+1|t} = \alpha_4^2\omega_{t|t} + R \tag{77}$$

which by substitution gives:

$$\omega_{t+1|t+1} = \omega_{t|t} - \frac{\omega_{t|t}^2 \alpha_4^2}{\alpha_4^2 \omega_{t|t} + R} + H \tag{78}$$

In steady state, $\omega_{t+1|t+1} = \omega_{t|t} = \omega_{00}$ which gives

$$\omega_{00} = \omega_{00} - \frac{\omega_{00}^2 \alpha_4^2}{\alpha_4^2 \omega_{00} + R} + H \tag{79}$$

Next, we solve for ω_{00}:

$$\frac{\omega_{00}^2 \alpha_4^2}{\alpha_4^2 \omega_{00} + R} - H = 0 \tag{80}$$

$$\frac{\omega_{00}^2}{\omega_{00} + R/\alpha_4^2} - H = 0 \tag{81}$$

$$\omega_{00}^2 - H(\omega_{00} + \frac{R}{\alpha_4^2}) = 0 \tag{82}$$

$$\omega_{00}^2 - H\omega_{00} - \frac{RH}{\alpha_4^2} = 0 \tag{83}$$

Set $\tau = \frac{R}{H\alpha_4^2}$:

$$\omega_{00}^2 - H\omega_{00} - H^2\tau = 0 \tag{84}$$

Solve with quadratic formula:

$$\frac{H \pm \sqrt{H^2 + 4H^2\tau}}{2} \tag{85}$$

$$\frac{H}{2} \pm \frac{1}{2}\sqrt{H^2(1 + 4\tau)} \tag{86}$$

$$\frac{H}{2} \pm \frac{H}{2}\sqrt{1 + 4\tau} \tag{87}$$

$$\frac{H}{2}(1 \pm \sqrt{1 + 4\tau}) \tag{88}$$

Replacing $\pm$ with $+$ gives

$$\omega_{00} = \frac{H}{2}(1 + \sqrt{1 + 4\tau}) \; where \; \tau = \frac{R}{H\alpha_4^2} \tag{89}$$

which is the final steady state conditional variance.

B.4 Derivation of Stationary Properties

The equation

$$Q^*_{t+1|t+1} = (R/\Sigma)^t Q^*_{0|1} + K\sum_{i=0}^{t-1}(R/\Sigma)^i(y_{t+1-i} - \delta x_{t-i}) \text{ for t} \geqslant 1 \tag{90}$$

shows that forcing the AR(1) coefficient on Q^*_t to equal one does not restrict the predicted state to be nonstationary. Instead, $Q^*_{t|t}$ is nonstationary iff $(y_{t+1-i} - \delta x_{t-i})$ is nonstationary because $(R/\Sigma)^i$ is absolutely summable, where x_t is a vector of covariates consisting of T^*_t and E^*_t and δ is the vector of coefficients on T^*_t and E^*_t in the measurement equation. This derivation is based on Lemma 3.2.2 of Miller (2005) and is given below.

Using the prediction and updating equations from section B.2, we get

$$Q^*_{t+1|t+1} = Q^*_{t|t} + K(y_{t+1} - y_{t+1|t}) \tag{91}$$

and by substitution of $y_{t+1|t}$ we obtain

$$Q^*_{t+1|t+1} = Q^*_{t|t} + K(y_{t+1} - [\alpha_4 Q^*_{t|t} + \delta x_t]) \tag{92}$$

Combining terms gives

$$Q^*_{t+1|t+1} = \frac{R}{\Sigma}Q^*_{t|t} + K(y_{t+1} - \delta x_t) \tag{93}$$

and substituting this equation for $Q^*_{t|t}$ iteratively gives

$$Q^*_{t+1|t+1} = (R/\Sigma)^t Q^*_{0|1} + K\sum_{i=0}^{t-1}(R/\Sigma)^i(y_{t+1-i} - \delta x_{t-i}) \text{ for t} \geqslant 1 \tag{94}$$

which completes most of the derivation. Replacing $\Sigma = \alpha_4^2\omega_{00}$ gives the final form

$$Q^*_{t+1|t+1} = (R/\Sigma)^t Q^*_{0|1} + K\sum_{i=0}^{t-1}(\frac{R}{R+\alpha_4^2\omega_{00}})^i(y_{t+1-i} - \delta x_{t-i}) \text{ for t} \geqslant 1 \tag{95}$$

which shows that $\frac{R}{R+\alpha_4^2\omega_{00}} < 1$ and therefore makes $(\frac{R}{R+\alpha_4^2\omega_{00}})^i$ absolutely summable.